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Modular fork assembly for a forked material-handling vehicle

Abstract

A modular fork assembly for a pallet truck includes a discrete elongate body, a discrete load wheel module, and a discrete fork tip that may be detachably connected by the same or different types of interlocking mechanisms that are adapted to inhibit shear forces. The load wheel module includes a frame and a load wheel assembly that includes a load wheel that is operatively connected to the frame.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION (1) This application is a continuation of U.S. patent application Ser. No. 16/656,854, entitled “Load Wheel Designs for Pallet Entry,” filed Oct. 18, 2019, the entire disclosure of which is incorporated by reference herein for all purposes. This application is also a continuation-in-part of U.S. patent application Ser. No. 16/367,050, entitled “Modular Fork Assembly for a Material-Handling Vehicle,” filed Mar. 27, 2019, the entire disclosure of which is incorporated by reference herein for all purposes.

TECHNICAL FIELD

(1) The field of this disclosure relates generally to forked vehicles configured to transport goods and materials. More specifically, this disclosure relates to fork assemblies for pallet trucks.

BACKGROUND INFORMATION

(2) Material-handling vehicles, such as forked vehicles, are presented herein only by way of example to pallet trucks. Typical pallet trucks support one, two in-line, or three in-line standard size pallets. Typically, pallet trucks include lifting load forks that are welded at their rear end or heel end to a chassis or battery box. The forks typically include an elongate body welded to elongate steel tubes to provide support. The front end of the forks typically includes support rollers. A hydraulic system, which is located in proximity to the chassis or battery box, operates a lifting mechanism that moves the support rollers, and lifts the chassis or battery box and the forks together with goods, such as pallets loaded thereon. The support rollers are typically coupled to the lift mechanism by a mechanical linkage that transmits force from a hydraulic lifting cylinder to the

support rollers. A valve arrangement is provided to relieve the hydraulic pressure in the lifting cylinder, thus lowering and placing the load on the floor. Steer wheels are located behind the battery box. A steering mechanism, such as a tiller, also may be provided to steer the steer wheels relative to the chassis and forks.

OVERVIEW OF DISCLOSURE

(3) This overview is provided to introduce a selection of concepts in a simplified form that are further described in greater detail later. This overview is not intended to identify key or essential inventive concepts of the claimed subject matter, nor is it intended for limiting the scope of the claimed subject matter.

(4) One aspect of this disclosure relates to fork assemblies that include modular detachable components.

(5) Another aspect of this disclosure relates to load wheel modules that contain a hydraulic actuator.

(6) In one embodiment, a fork assembly for a forked material-handling truck (e.g., pallet truck) comprises a discrete elongate body; a discrete load wheel module; a first interlocking mechanism configured for detachable connection of the elongate body to the load wheel module; a discrete fork tip; and a second interlocking mechanism configured for detachable connection of the load wheel module to the fork tip.

(7) In some additional, alternative, or selectively cumulative embodiments, a load wheel module for a fork assembly for a forked material-handling truck (e.g., pallet truck) comprises a frame; a load wheel assembly, including a load wheel, operatively connected to the frame; and a hydraulic actuator contained within the frame and operatively connected to the load wheel assembly to lower the load wheel hydraulically.

(8) In some additional, alternative, or selectively cumulative embodiments, an inventory of parts for a fork assembly for a forked material-handling truck (e.g., pallet truck) comprises multiple interchangeable elongate bodies; multiple interchangeable load wheel modules; multiple interchangeable fork tips; and multiple operatively identical interlocking mechanism components configured for detachable connection of any one of the elongate bodies to any one of the load wheel modules and configured for detachable connection of any one of the fork tips to any one of the load wheel modules.

(9) In some additional, alternative, or selectively cumulative embodiments, a forked material-handling truck (e.g., pallet truck) including a fork assembly comprises a steer wheel; a chassis operatively connected to the steer wheel; and two substantially parallel forks operatively connected to and extending from the chassis and configured to hold a load for conveyance by the pallet truck as the pallet truck moves, the forks including a first fork and a second fork, wherein the first fork comprises a first elongate body, a first load wheel module, a first interlocking mechanism configured for detachable connection of the first elongate body to the first load wheel module, a first fork tip, and a second interlocking mechanism configured for detachable connection of the first load wheel module to the first fork tip, wherein the second fork comprises a second elongate body, a second load wheel module, a third interlocking mechanism configured for detachable connection of the second elongate body to the second load wheel module, a second fork tip, and a fourth interlocking mechanism configured for detachable connection of the second load wheel module to the second fork tip.

(10) In some additional, alternative, or selectively cumulative embodiments, a method for deploying a load wheel of a forked material-handling truck (e.g., pallet truck) in response to a load wheel deployment signal comprises providing hydraulic power through a hydraulic line positioned within an elongate body of a fork assembly; employing the hydraulic power from the hydraulic line to actuate multiple pistons of an actuator positioned within a load wheel module of the fork assembly, wherein the load wheel module has a load wheel module frame that is operatively connected to the elongate body; and employing the multiple pistons to deploy the load wheel so that it is vertically spaced apart from the load wheel module frame.

- (11) In some additional, alternative, or selectively cumulative embodiments, a method of assembling a fork for a forked material-handling vehicle comprises detachably connecting a modular elongate body to a modular load wheel module; and detachably connecting the modular load wheel module to a modular fork tip.
- (12) In some additional, alternative, or selectively cumulative embodiments, a fork for a forked material-handling vehicle comprises a frame; a wheel operatively connected to the frame; and a hydraulic actuator contained within the frame and operatively connected to the wheel to lower the wheel hydraulically, the hydraulic actuator having an input for connection to a hydraulic hose running along a portion of the length of the fork from a hydraulic pressure source.
- (13) In some additional, alternative, or selectively cumulative embodiments, the first interlocking mechanism and the second interlocking mechanism are substantially identical.
- (14) In some additional, alternative, or selectively cumulative embodiments, the first interlocking mechanism and the second interlocking mechanism contain a substantially identical component.
- (15) In some additional, alternative, or selectively cumulative embodiments, the first interlocking mechanism and the second interlocking mechanism are operatively identical.
- (16) In some additional, alternative, or selectively cumulative embodiments, the first interlocking mechanism and the second interlocking mechanism are interchangeable.
- (17) In some additional, alternative, or selectively cumulative embodiments, the first interlocking mechanism and the second interlocking mechanism contain an interchangeable component.
- (18) In some additional, alternative, or selectively cumulative embodiments, the first and second load wheel modules are interchangeable, the first and second tips are interchangeable, the first and third interlocking mechanisms are operatively identical, and the second and fourth interlocking mechanisms are operatively identical.
- (19) In some additional, alternative, or selectively cumulative embodiments, the first interlocking mechanism and the second interlocking mechanism contain a reusable component.
- (20) In some additional, alternative, or selectively cumulative embodiments, at least one of the first interlocking mechanism and the second interlocking mechanism employs one or more mated pairs of internally threaded receptacles and externally threaded fasteners.
- (21) In some additional, alternative, or selectively cumulative embodiments, the externally threaded fasteners have a minimum diameter of about 12 mm.
- (22) In some additional, alternative, or selectively cumulative embodiments, at least one of the first interlocking mechanism and the second interlocking mechanism employs mated shear-resistant features, including first and second shear-resistant features.
- (23) In some additional, alternative, or selectively cumulative embodiments, the first shear-resistant feature includes a protruding feature and the second shear-resistant feature includes a receiving feature.
- (24) In some additional, alternative, or selectively cumulative embodiments, the first interlocking mechanism includes a first protruding feature that is mated to a first receiving feature, wherein one of the first protruding feature and the first receiving feature is permanently connected to or associated with the elongate body, wherein a different one of the first protruding feature and the first receiving feature is permanently connected to or associated with the load wheel module.
- (25) In some additional, alternative, or selectively cumulative embodiments, the shear-resistant features are configured to receive a fastener.
- (26) In some additional, alternative, or selectively cumulative embodiments, the first interlocking mechanism employs a first internally threaded receptacle that is mated to a first externally threaded fastener, wherein one of the first internally threaded receptacle and first externally threaded fastener is configured to connect to the elongate body, and wherein a different one of the first internally threaded receptacle and the first externally threaded fastener is configured to connect to the load wheel module.
- (27) In some additional, alternative, or selectively cumulative embodiments, both of the first

protruding feature and the first receiving feature are configured to receive the first externally threaded fastener.

(28) In some additional, alternative, or selectively cumulative embodiments, the first interlocking mechanism includes a first endcap that is attached to the elongate body and includes a first shear-resistant body feature that is mated to a first shear-resistant module feature of the load wheel module.

(29) In some additional, alternative, or selectively cumulative embodiments, the first shear-resistant body feature and the first shear-resistant module feature are adapted to receive a fastener.

(30) In some additional, alternative, or selectively cumulative embodiments, the first endcap is welded to the elongate body.

(31) In some additional, alternative, or selectively cumulative embodiments, the second interlocking mechanism includes a second endcap that is attached to the fork tip and includes a second shear-resistant attachment feature that is mated to a second shear-resistant module feature of the load wheel module, wherein the first and second interlocking mechanisms are operatively identical.

(32) In some additional, alternative, or selectively cumulative embodiments, the second interlocking mechanism includes a second endcap that is attached to the fork tip and includes a second shear-resistant attachment feature that is mated to a second shear-resistant module feature of the load wheel module.

(33) In some additional, alternative, or selectively cumulative embodiments, the second shear-resistant attachment feature and the second shear-resistant module feature are adapted to receive a fastener.

(34) In some additional, alternative, or selectively cumulative embodiments, the elongate body has a first characterizing color, wherein the load wheel module has a second characterizing color, wherein the fork tip has a third characterizing color, and wherein the first, second, and third characterizing colors are different.

(35) In some additional, alternative, or selectively cumulative embodiments, the load wheel module has opposing ends having substantially identical shear-resistant features.

(36) In some additional, alternative, or selectively cumulative embodiments, the elongate body includes a channel along its length, and wherein an endcap associated with the first interlocking mechanism includes an aperture that aligns with the channel.

(37) In some additional, alternative, or selectively cumulative embodiments, the fork tip includes a proximal connection end for attachment closest to the load wheel module, wherein the proximal end has proximal end dimensions, wherein the fork tip includes a distal end that has distal end dimensions, and wherein at least one of the distal end dimensions is smaller than a respective one of the proximal end dimensions.

(38) In some additional, alternative, or selectively cumulative embodiments, the pallet truck comprises: a hydraulic power source; and a first hydraulic line positioned through the first elongate body, wherein the first hydraulic line transmits hydraulic fluid from the hydraulic power source to a first hydraulic actuator positioned within the first load wheel module.

(39) In some additional, alternative, or selectively cumulative embodiments, the load wheel module comprises: a frame; a load wheel assembly, including a load wheel, operatively connected to the frame; and a hydraulic actuator contained within the frame and operatively connected to the load wheel assembly to lower the load wheel hydraulically.

(40) In some additional, alternative, or selectively cumulative embodiments, the load wheel module comprises a frame; a load wheel assembly, including a load wheel, operatively connected to the frame; and a mechanical link operatively coupled to the load wheel assembly to lower the load wheel, wherein the mechanical link extends through the discrete elongate body.

(41) In some additional, alternative, or selectively cumulative embodiments, the pallet truck comprises a power source and a first mechanical link positioned through the first elongate body,

wherein the first mechanical link transmits force from the power source to a first load wheel mechanism positioned within the first load wheel module.

(42) In some additional, alternative, or selectively cumulative embodiments, the pallet truck comprises: first and second elongate bodies having a first characterizing color, first and second load wheel modules having a second characterizing color, first and second fork tips having a third characterizing color, wherein the first, second, and third characterizing colors are different.

(43) In some additional, alternative, or selectively cumulative embodiments, the hydraulic actuator includes a hydraulic line input connector operative for connecting the hydraulic actuator to a hydraulic line that transmits hydraulic fluid from a hydraulic power source.

(44) In some additional, alternative, or selectively cumulative embodiments, hydraulic power from the hydraulic power source has a maximum pressure within the range of about **2,000** to about 4,000 psi at the hydraulic line input connector.

(45) In some additional, alternative, or selectively cumulative embodiments, the hydraulic actuator includes multiple pistons.

(46) In some additional, alternative, or selectively cumulative embodiments, the multiple pistons are hydraulically connected to a hydraulic manifold.

(47) In some additional, alternative, or selectively cumulative embodiments, the hydraulic manifold is hydraulically connected to a hydraulic line input connector operative for connecting the hydraulic actuator to a hydraulic line that transmits hydraulic fluid from a hydraulic power source.

(48) In some additional, alternative, or selectively cumulative embodiments, at least one of the pistons has a length within the range of about 0.50 inches to about 3 inches (about 1.27 to about 7.62 cm) and a stroke length within the range of about 1 to about 3 inches (about 2.54 to about 7.62 cm).

(49) In some additional, alternative, or selectively cumulative embodiments, at least one of the pistons has a length within the range of about 1 to about 2 inches (about 2.54 to about 5.08 cm) and a stroke length within the range of about 1.5 to about 2.5 inches (about 3.81 to about 6.35 cm).

(50) In some additional, alternative, or selectively cumulative embodiments, the hydraulic actuator includes at least three pistons.

(51) In some additional, alternative, or selectively cumulative embodiments, the hydraulic actuator includes at least four pistons.

(52) In some additional, alternative, or selectively cumulative embodiments, the hydraulic actuator is operable to provide maximum thrust within a range of about 66,723 newtons (about 15,000 pounds) to about 133,446 newtons (about 30,000 pounds).

(53) In some additional, alternative, or selectively cumulative embodiments, the hydraulic actuator is operable to provide greater than about 66,723 newtons (about 15,000 pounds) of thrust.

(54) In some additional, alternative, or selectively cumulative embodiments, the hydraulic actuator is operable to provide greater than about 88,964 newtons (about 20,000 pounds) of thrust.

(55) In some additional, alternative, or selectively cumulative embodiments, the hydraulic actuator is operatively connected to the load wheel module frame.

(56) In some additional, alternative, or selectively cumulative embodiments, the frame includes a fork tip-facing end and a body-facing end, and wherein the actuator is positioned closer to the body-facing end than to the fork tip-facing end.

(57) In some additional, alternative, or selectively cumulative embodiments, the body-facing end includes an aperture adapted to accommodate a hydraulic line that transmits hydraulic fluid from a hydraulic power source to the hydraulic actuator.

(58) In some additional, alternative, or selectively cumulative embodiments, the load wheel module comprises: a first attachment feature of a first interlocking mechanism configured for detachable connection of the body-facing end of the frame to an elongate body; and a second attachment feature of a second interlocking mechanism configured for detachable connection of the fork tip-facing end of the frame to a fork tip.

- (59) In some additional, alternative, or selectively cumulative embodiments, at least one of the first and second attachment features includes a sheer-resistant module feature that is mated to a sheer-resistant feature of the elongate body or the fork tip.
- (60) In some additional, alternative, or selectively cumulative embodiments, the sheer-resistant module feature includes one or more receiving features.
- (61) In some additional, alternative, or selectively cumulative embodiments, the load wheel module frame includes a fork tip-facing end and a body-facing end, wherein the load wheel is positioned closer to the fork tip-facing end than to the body-facing end.
- (62) In some additional, alternative, or selectively cumulative embodiments, the load wheel is one of multiple load wheels that are part of the load wheel module.
- (63) In some additional, alternative, or selectively cumulative embodiments, the hydraulic actuator and the load wheel assembly form a wheel module substructure, wherein the load wheel assembly includes a load wheel unit and a wheel carrier strut that are operatively connected to each other, and wherein the wheel carrier strut is operatively connected to hydraulic actuator and the frame.
- (64) In some additional, alternative, or selectively cumulative embodiments, the wheel carrier strut is pivotally connected to hydraulic actuator and pivotally connected to the frame, and wherein the wheel carrier strut is pivotally connected to the load wheel unit.
- (65) Selectively cumulative embodiments are embodiments that include any combination of multiple embodiments that are not mutually exclusive.
- (66) Additional aspects and advantages will be apparent from the following detailed description of example embodiments, which proceeds with reference to the accompanying drawings.
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Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1A is an orthogonal view of an example of a prior-art closed-bottom pallet.
- (2) FIG. 1B illustrates a front left isometric view of a prior-art lift fork assembly, showing a pair of forks welded to a battery box.
- (3) FIG. 1C is a top schematic view of a prior-art pallet truck.
- (4) FIG. 1D illustrates a cross-sectional view of a prior-art elongate body of a lift fork.
- (5) FIG. 1E is an isometric view of a load wheel unit of a prior-art pallet truck.
- (6) FIGS. 1F-1I are a series of side elevation views showing a progression of a fork approaching a closed-bottom pallet, encountering the closed-bottom pallet, negotiating a base board of the closed-bottom pallet, and deploying a load wheel extension mechanism to lift the closed-bottom pallet off of a supporting surface.
- (7) FIG. 2 illustrates a front left partly exploded isometric view of an example battery box and a fork assembly showing the fork assembly disassembled from the battery box, according one embodiment.
- (8) FIG. 3A illustrates a top plan view of a fork assembly, according one embodiment.
- (9) FIG. 3B illustrates a right-side elevation view of the fork assembly shown in FIG. 3A.
- (10) FIG. 3C illustrates a bottom view of the fork assembly shown in FIG. 3A.
- (11) FIG. 3D illustrates a bottom right isometric view of a portion of the fork assembly, showing components of an embodiment of a first interlocking mechanism.
- (12) FIG. 4A illustrates a front right partly exploded isometric view of a portion of the fork assembly shown in FIG. 3A.
- (13) FIG. 4B illustrates a front right exploded isometric view of a portion of the fork assembly, showing components of an embodiment of a first interlocking mechanism.
- (14) FIG. 4C illustrates a front right exploded isometric view of a portion of the fork assembly, showing components of an embodiment of a second interlocking mechanism.

- (15) FIG. 4D illustrates a front right isometric view of an embodiment of an endcap.
- (16) FIG. 4E illustrates a bottom rear right isometric view of an embodiment of an endcap.
- (17) FIG. 4F illustrates a rear left bottom isometric view of a body-facing end of a load wheel module, according to one embodiment.
- (18) FIG. 5A illustrates a front right isometric view of an elongate body of the fork assembly shown in FIG. 3A.
- (19) FIG. 5B illustrates a cross-sectional view of the elongate body shown in FIG. 5A.
- (20) FIG. 6A illustrates a front left isometric view of an elongate body, according to an alternative embodiment.
- (21) FIG. 6B illustrates a cross-sectional view of the elongate body shown in FIG. 6A.
- (22) FIG. 6C illustrates a front left isometric view of an elongate body, according to another alternative embodiment.
- (23) FIG. 7 illustrates a cross-sectional view of an elongate body portion, according to another alternative embodiment.
- (24) FIG. 8A illustrates a front right bottom isometric view of a portion of an elongate body showing an endcap, according to one embodiment.
- (25) FIG. 8B illustrates a top front right isometric view of an elongate body of a fork assembly, wherein the elongate body is attached to endcaps, according to one embodiment.
- (26) FIG. 8C illustrates a top front right enlarged isometric view of a portion of an elongate body showing an endcap, according to one embodiment.
- (27) FIG. 9A illustrates a front right bottom isometric view of a load wheel module of a fork assembly, showing an undeployed load wheel unit, according to one embodiment.
- (28) FIG. 9B illustrates a front right bottom isometric view of a load wheel module of a fork assembly, showing a deployed load wheel unit, according to one embodiment.
- (29) FIG. 9C illustrates a front right top isometric view of a wheel module substructure of a load wheel module, according to one embodiment.
- (30) FIG. 10A is a top plan view of a load wheel assembly employing an embodiment of a torque-coupling assembly.
- (31) FIG. 10B is a side elevation view of the load wheel assembly of FIG. 10A with a wheel carrier strut and a portion of a wheel carrier bracket removed for clarity.
- (32) FIG. 11A is a top plan view of a load wheel assembly employing an alternative embodiment of a torque-coupling assembly.
- (33) FIG. 11B is a side elevation view of the load wheel assembly of FIG. 11A with a wheel carrier strut and a portion of a wheel carrier bracket removed for clarity.
- (34) FIG. 11C shows a side elevation view of a leaf spring of FIG. 11B in one example of a relaxed shape when the leaf spring is not yet deployed over an idler axle and connected to a wheel carrier.
- (35) FIG. 12A is a top plan view of a load wheel assembly employing an alternative embodiment of a torque-coupling assembly.
- (36) FIG. 12B is a side elevation view of the load wheel assembly of FIG. 12A with a wheel carrier strut and a portion of a wheel carrier bracket removed for clarity.
- (37) FIG. 12C is a side elevation view of an idler wheel of the torque-coupling assembly shown in FIG. 12A.
- (38) FIG. 13A is a top plan view of a load wheel assembly employing another alternative embodiment of a torque-coupling assembly.
- (39) FIG. 13B is a side elevation view of the load wheel assembly of FIG. 13A with a wheel carrier strut and a portion of a wheel carrier bracket removed for clarity.
- (40) FIG. 13C shows a side elevation view of the torsion clip of FIG. 13B in one example of a relaxed shape when the torsion clip is not yet deployed over an idler axle.
- (41) FIG. 14A is a top plan view of a load wheel assembly employing another alternative embodiment of a torque-coupling assembly.

- (42) FIG. 14B is a side elevation view of the load wheel assembly of FIG. 14A with a wheel carrier strut and a portion of a wheel carrier bracket removed for clarity.
- (43) FIG. 14C shows a side elevation view of the torsion clip of FIG. 14B in one example of a relaxed shape when the torsion clip is not yet deployed about the ends of the wheel carrier.
- (44) FIG. 15A is a top plan view of a load wheel assembly employing another alternative embodiment of a torque-coupling assembly.
- (45) FIG. 15B is a side elevation view of the load wheel assembly of FIG. 15A with a wheel carrier strut and a portion of a wheel carrier bracket removed for clarity.
- (46) FIG. 16A is a top plan view of a load wheel assembly employing another alternative embodiment of a torque-coupling assembly.
- (47) FIG. 16B is a side elevation view of the load wheel assembly of FIG. 16A with a wheel carrier strut and a portion of a wheel carrier bracket removed for clarity.
- (48) FIG. 17A is a top plan view of a load wheel assembly employing another alternative embodiment of a torque-coupling assembly.
- (49) FIG. 17B is a side elevation view of the load wheel assembly of FIG. 17A with a wheel carrier strut and a portion of a wheel carrier bracket removed for clarity.
- (50) FIG. 18A is a top plan view of a load wheel assembly employing another alternative embodiment of a torque-coupling assembly with a force-applying coupling assembly positioned outside the wheel carrier.
- (51) FIG. 18B is a side elevation view of the load wheel assembly of FIG. 18A with a wheel carrier strut and a portion of a wheel carrier bracket removed for clarity.
- (52) FIG. 18C is a top plan view of a load wheel assembly employing another alternative embodiment of a torque-coupling assembly with a force-applying coupling assembly positioned inside the wheel carrier.
- (53) FIG. 18D is a side elevation view of the load wheel assembly of FIG. 18C with a wheel carrier strut and a portion of a wheel carrier bracket removed for clarity.
- (54) FIG. 19A is a top plan view of a load wheel assembly employing another alternative embodiment of a torque-coupling assembly.
- (55) FIG. 19B is a side elevation view of the load wheel assembly of FIG. 19A with a wheel carrier strut and a portion of a wheel carrier bracket removed for clarity.
- (56) FIG. 20A is a top plan view of a load wheel assembly employing another alternative embodiment of a torque-coupling assembly.
- (57) FIG. 20B is a side elevation view of the load wheel assembly of FIG. 20A with a wheel carrier strut and a portion of a wheel carrier bracket removed for clarity.
- (58) FIG. 21A is a top plan view of a load wheel assembly employing another alternative embodiment of a torque-coupling assembly.
- (59) FIG. 21B is a side elevation view of the load wheel assembly of FIG. 21A with a wheel carrier strut and a portion of a wheel carrier bracket removed for clarity.
- (60) FIG. 22A is a top plan view of a load wheel assembly employing another alternative embodiment of a torque-coupling assembly.
- (61) FIG. 22B is a side elevation view of the load wheel assembly of FIG. 22A with a wheel carrier strut and a portion of a wheel carrier bracket removed for clarity.
- (62) FIG. 22C is a top plan view of a load wheel assembly similar to that shown in FIG. 22A with an added lead-in roller in the form of a paddle wheel.
- (63) FIG. 22D is a first side elevation view of the load wheel assembly of FIG. 22D with a wheel carrier strut and a portion of a wheel carrier bracket removed for clarity.
- (64) FIG. 22E is a second side elevation view of the load wheel assembly of FIG. 22D with a wheel carrier strut and a portion of a wheel carrier bracket removed for clarity.

DETAILED DESCRIPTION OF EMBODIMENTS

- (65) Example embodiments are described below with reference to the accompanying drawings.

Unless otherwise expressly stated in the drawings, the sizes, positions, etc., of components, features, elements, etc., as well as any distances therebetween, are not necessarily to scale, and may be disproportionate and/or exaggerated for clarity.

(66) The terminology used herein is for the purpose of describing particular example embodiments only and is not intended to be limiting. As used herein, the singular forms “a,” “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It should be recognized that the terms “comprise,” “comprises,” and/or “comprising,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof. Unless otherwise specified, a range of values, when recited, includes both the upper and lower limits of the range, as well as any sub-ranges therebetween. Unless indicated otherwise, terms such as “first,” “second,” etc., are only used to distinguish one element from another. For example, one element could be termed a “first element” and similarly, another element could be termed a “second element,” or vice versa. The section headings used herein are for organizational purposes only and are not to be construed as limiting the subject matter described.

(67) Unless indicated otherwise, the terms “about,” “thereabout,” “substantially,” etc. mean that amounts, sizes, formulations, parameters, and other quantities and characteristics are not and need not be exact, but may be approximate and/or larger or smaller, as desired, reflecting tolerances, conversion factors, rounding off, measurement error and the like, and other factors known to those of skill in the art.

(68) Spatially relative terms, such as “right,” “left,” “below,” “beneath,” “lower,” “above,” and “upper,” and the like, may be used herein for ease of description to describe one element's or feature's relationship to another element or feature, as illustrated in the drawings. It should be recognized that the spatially relative terms are intended to encompass different orientations in addition to the orientation depicted in the figures. For example, if an object in the figures is turned over, elements described as “below” or “beneath” other elements or features would then be oriented “above” the other elements or features. Thus, the term “below” can, for example, encompass both an orientation of above and below. An object may be otherwise oriented (e.g., rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may be interpreted accordingly.

(69) Unless clearly indicated otherwise, all connections and all operative connections may be direct or indirect. Similarly, unless clearly indicated otherwise, all connections and all operative connections may be rigid or non-rigid.

(70) Like numbers refer to like elements throughout. Thus, the same or similar numbers may be described with reference to other drawings even if they are neither mentioned nor described in the corresponding drawing. Also, even elements that are not denoted by reference numbers may be described with reference to other drawings.

(71) Many different forms and embodiments are possible without deviating from the spirit and teachings of this disclosure and so this disclosure should not be construed as limited to the example embodiments set forth herein. Rather, these example embodiments are provided so that this disclosure will be thorough and complete, and will convey the scope of the disclosure to those skilled in the art.

(72) FIG. 1A is an orthogonal view of an example of a closed-bottom pallet (also called a closed pallet) **1** that is commonly used in the UK, USA, and Australia. The closed pallet **1** has one or more base boards **2** that contact a supporting surface **3**, such as the floor. The base boards **2** provide the foundation for spacers **4** that support one or more upper planks **5** that support a load.

(73) FIG. 1B illustrates a front left isometric view of a prior-art fork and battery box assembly **32** having a pair of forks **33** welded to a battery box **36**, and FIG. 1D illustrates a cross-sectional view of a prior art elongate body of a lift fork **33**. As is typical with conventional pallet trucks, each of

the forks **33** is made of multiple components (such as a load-bearing surface **34** and support tubes **35**) connected by longitudinal welds **37** to form a unitary structure that is welded to the battery box **36** and to a torsion member **38**.

(74) FIG. **1C** is a top schematic view of a prior pallet truck **9**, which is an example of a forked material-handling vehicle. Pallet trucks **9**, such as pedestrian pallet trucks (PPT) or rider pallet trucks (RPT), are often employed to handle pallets that are on the ground or on another horizontal surface, for example, when loading or unloading from a transport vehicle, such as a truck or shipping container. Pallet trucks **9** typically include a chassis **10** that supports or is attached to a battery box **11**. The chassis **10** or the battery box **11** is connected to a pair of forks **12**, each fork **12** having an elongate body **13**, a forward load wheel **14**, a rear load wheel **15**, and a tip **16**. The forward load wheel **14** and the rear load wheel **15** can be collectively referred to as load wheels **17**.

(75) FIG. **1E** is an isometric view of a load wheel unit **20** of a prior art pallet truck **9**. FIGS. **1F-1I** are a series of side elevation views showing a progression of a fork **12** approaching a closed-bottom pallet **1** in a forward direction **19**, encountering the base board **2** of the closed-bottom pallet **1**, negotiating the base board **2** of the closed-bottom pallet **1**, and deploying a load wheel extension mechanism to lift the closed-bottom pallet **1** off of a supporting surface **3**. With reference to FIGS. **1A**, **1C**, **1E**, and **1F-1I**, when engaging or disengaging closed pallets **1**, the forward load wheel **14** and the rear load wheel **15** of the pallet truck **9** cross over the top of the base boards **2** of the closed pallet **1** to enter and exit pallet pockets **6**. When a closed pallet **1** is empty, or has a light-weight palletized load (commonly meaning bearing a load of 25 kg or less), the forward load wheel **14** of the pallet truck **9** may lock upon contact with the closed pallet **1** and stop rotation instead of riding over the base boards **2** and into the pallet pockets **6** of the closed pallet **1**. However, the rear load wheel **15** may continue to rotate so that the closed pallet **1** may be shunted or pushed horizontally across the surface **3**. Similarly, the rear load wheel **15** may lock while the forward load wheel continues to roll during attempts to exit the pallet pockets **6**. Such shunting is undesirable because it can damage the base board **2** and/or the surface **3** on which the closed pallet **1** sits and may move the pallet **1** to an undesired location. Also, failure to enter the pallet **1** prevents the pallet **1** from being retrieved, and failure to exit an empty pallet **1** renders the truck **9** useless with a pallet **1** trapped on the forks **12** until the pallet **1** is removed.

(76) One challenge faced by pallet truck manufacturers is that customers often want varying fork configurations, such as forks with variable spreads, lengths, tips, and widths. Because forks are typically manufactured in standard sizes, changing fork parameters requires costly and time-consuming retooling to modify the battery box and/or fork design to produce a pallet truck conforming to individual customer specifications. In some situations, such redesigns can add up to six weeks of lead-time. In addition, stocking multiple lengths of forks may require a significant capital outlay for inventory. Some of these issues can be addressed by the development of a modular fork assembly with detachably connectable components that couples directly or indirectly to the battery box, chassis, or other part connected to the chassis.

(77) FIG. **2** illustrates a front left isometric view of an example of a fork and battery box assembly **39** with a modular fork assembly **40** disassembled from a battery box **41** of a pallet truck, such as one of the HYSTER W45ZHD2 or YALE MPB045ZH manufactured by the Hyster-Yale Group, 5200 Martin Luther King Junior Highway, Greenville, NC 27834. A typical fork and battery box assembly **39** includes one battery box **41** and multiple modular fork assemblies **40** (such as two modular fork assemblies **40**), though only one modular fork assembly **40** is depicted in FIG. **2**. The modular fork assemblies **40** may be coupled to the battery box **41** by welding or by locking. Locking a modular fork assembly **40** to the battery box **41**, the torsion member **38**, or both, means that the modular fork assembly **40** can also be unlocked from the battery box **41**, the torsion member **38**, or both.

(78) The modular fork assembly **40** may be coupled to the battery box **41** through one or more intermediary couplers or interlocking mechanisms, such as an optional heel **45** shown in FIG. **2**.

The optional heel **45** can be connected to the elongate body **50**, for example, by welding or other suitable attachment, such as later described. A proximal end **42** (also referred to as a heel end) of the elongate body **50** is the end closest to the battery box **41**. As illustrated, the proximal end **42** of the modular fork assembly **40** is coupled to the heel **45**, which is configured to be locked to the battery box **41** and/or the torsion member **38**; however, the proximal end **42** of the modular fork assembly **40** may be welded or otherwise coupled directly to the battery box **41** and/or the torsion member **38** with or without employing a separate optional heel **45**.

(79) The battery box **41** is sized to fit a battery or battery array. When used in conjunction with a pallet truck, pallet jack, or other suitable forklift, the entire fork and battery box assembly **39** may be raised and lowered as a single unit, for example via a hydraulic cylinder actuated by hydraulic power from a hydraulic power source (not shown). The two modular fork assemblies **40** may be referred to as a right modular fork assembly and a left modular fork assembly, depending on the side of the battery box **41** to which they are coupled. These right and left modular fork assemblies may be identical such that one modular fork assembly may be swapped for the other.

(80) FIG. 3A, FIG. 3B, and FIG. 3C (collectively FIG. 3) illustrate respective top plan, right side elevation, bottom, and bottom right isometric views of an example of a modular fork assembly **40**. FIG. 4A illustrates a front right partly exploded isometric view of a portion of the modular fork assembly **40** shown in FIG. 3A; FIG. 4B illustrates a front right exploded isometric view of a portion of the modular fork assembly **40**, showing components of an embodiment of a first interlocking mechanism **365**; FIG. 4C illustrates a front right exploded isometric view of a portion of the modular fork assembly, showing components of an embodiment of a second interlocking mechanism **675**; FIG. 4D illustrates a front right isometric view of an embodiment of an endcap **360**; FIG. 4E illustrates a bottom rear right isometric view of an embodiment of the endcap **360**; and FIG. 4F illustrates a rear left bottom isometric view of a body-facing end **65** of the load wheel module **55**.

(81) With reference to FIG. 3 and FIGS. 4A, 4B, 4C, 4D, 4E, and 4F (collectively FIG. 4) each modular fork assembly **40** includes multiple components. A fully assembled modular fork assembly **40** includes an elongate body **50**, a load wheel module **55**, and a fork tip **60** (also referred to as a fork toe). A proximal or body-facing end **65** of the load wheel module **55** can be detachably connected to a distal end **70** (opposite the proximal end **42**) of the elongate body **50**, the distal end **70** being furthest from the battery box **41**. And, the fork tip **60** can be detachably connected to a distal or fork tip-facing end **75** of the load wheel module **55**. The elongate body **50**, the load wheel module **55**, and the fork tip **60** may be randomly selected from an inventory of respective substantially identical elongate bodies **50**, load wheel modules **55**, and fork tips **60**. To facilitate ease of storage and assembly, the elongate bodies **50** may have a first characterizing color; the load wheel modules **55** may have a second characterizing color; and the fork tips **60** may have a third characterizing color, wherein the first, second, and third characterizing colors are different.

(82) For convenience and modularity, the optional heel **45**, the elongate body **50**, and the fork tip **60** may be identical for both the left and right modular fork assemblies **40** (e.g. the fork assemblies **40** coupled to the left and right sides of the battery box **41**). Using identical components for both the left and right fork assemblies **40** increases the modularity of the system over a system in which the left and right forks are made with distinct, non-interchangeable components. However, distinct, non-identical exchangeable components may be used to create different left and right modular fork assemblies **40**. For example, the elongate body **50** may be made in any desired length, detachably coupled to one of several different designs for the load wheel module **55**, which is in turn detachably coupled to a desired fork tip **60** to create a customizable modular fork assembly **40** to accommodate a wide range of customer preferences.

(83) Although a fork body of conventional cross-sectional construction can be employed, the elongate body **50** can alternatively be formed such that thinner and/or lighter materials may be used compared to existing fork bodies while providing excellent dimensional stability and reducing

materials costs and/or weight. In some embodiments, the elongate body **50** may include no more than two longitudinal weldments, or no more than one longitudinal weldment. In some embodiments, the method further includes forming a longitudinal weldment to join the first longitudinal edge and the second longitudinal edge of the steel sheet; and, in particular embodiments, the longitudinal weldment may extend the full length of the elongate body portion.

(84) Additionally, manufacturing processes that avoid the need to separately weld multiple parts together may be used, such as roll forming, additive manufacturing, or extrusion processes. One method of making the elongate body **50** for the modular fork assembly **40** includes using a rolling process to form a steel sheet into the elongate body **50**, which may include an understructure **80** and a load-bearing upper structure **85**. The rolling process may be a cold rolling process wherein the shape of the understructure is designed to be integrally strong so as to be substantially weldless or employ a few welds out of an abundance of caution. Alternatively a hot or warm rolling process may provide self-melding between components of the elongate body **50** that touch each other. The rolling process can be a continuous process that forms all the components of a cross section of the elongate body **50**. Another method includes using an extrusion process to form the elongate body portion **50**. The optional use of such processes to form the elongate body **50** reduces the assembly and welding costs typically associated with conventional fork manufacture. These and other features provide a competitive advantage and differentiator in an exceedingly crowded market.

(85) FIG. 5A and FIG. 5B (collectively FIG. 5) illustrate a front right isometric view and a cross-sectional view, respectively, of an example of an elongate body **50** of a fork assembly **40**. With reference to FIG. 5, the elongate body **50** may include an exemplary understructure **80** coupled to a load-bearing upper structure **85**, which may form an upper surface **115** of the elongate body **50**. The understructure **80** forms a structural element that resists one or more of flex, torsion, axial compression, and/or lateral deflection of the load-bearing upper structure **85**. The understructure **80** may include a first truss **90** and a second truss **95**. The first truss **90** may include a first strut **100** that extends downward from the outer edge of the load-bearing upper structure **85** in a generally orthogonal orientation with respect to the load-bearing upper structure **85**. A first cross beam **105** is coupled to the first strut **100** and extends away from the first strut **100**, for example, substantially orthogonally from the first strut **100** (toward a longitudinal midline **110** of the elongate body **50**) to form a lower surface **120** of the elongated body **50**. A second strut **125** may be coupled to the first cross beam **105** and may extend from the first cross beam **105** toward the load-bearing upper structure **85**.

(86) A second cross beam **130** may be coupled to the second strut **125** and may contact the lower surface **135** of the load-bearing upper structure **85**. The second cross beam **130** may be optionally coupled to the load-bearing upper structure **85**, for example, via spot welds or by being integrally formed with the load-bearing upper structure **85**. A third strut **140** is coupled to the second cross beam **130** and extends from the second cross beam **130** away from the load-bearing upper structure **85**. A third cross beam **145** is coupled to the third strut **140** and extends away from the third strut **140** (toward the midline **110** of the elongate body **50**) to form another lower surface **120** of the elongate body **50**. A fourth strut **150** extends from the third cross beam **145** towards the load-bearing upper structure **85** and may be coupled to the load-bearing upper structure **85**.

(87) The second truss **95** comprises a fifth strut **155** that extends downward from the load-bearing upper structure **85** and may be coupled to the load-bearing upper structure **85**. A fourth cross beam **160** is coupled to the fifth strut **155** and extends away from the fifth strut **155**, for example, substantially orthogonally from the fifth strut **204** (away from the longitudinal midline **110** of the elongate body **50**) to form another lower surface **120** of the elongated body **50**. A sixth strut **165** is coupled to the fourth cross beam **160** and extends from the fourth cross beam **165** towards the load-bearing upper structure **85**.

(88) A fifth cross beam **170** is coupled to the sixth strut **165** and contacts the lower surface **135** of the load-bearing upper structure **85**. The fifth cross beam **170** is optionally coupled to the load-

bearing upper structure **85**, for example, via spot welds or by being integrally formed with the load-bearing upper structure **85**. A seventh strut **175** is coupled to the fifth cross beam **170** and extends from the fifth cross beam **170** away from the load-bearing upper structure **85**. A sixth cross beam **180** is coupled to the seventh strut **175** and extends away from the seventh strut **175** (away from the midline **110** of the elongate body **150**) to form another lower surface **120** of the elongate body **50**. An eighth strut **185** extends from the sixth cross beam **180** towards the load-bearing upper structure **85** and may be coupled to the load-bearing upper structure **85**. The struts may be generally vertical, or one or more of the struts may be angled with respect to the plane(s) of first and second surface zones **190** and **195** of the load-bearing upper structure **85**.

(89) The load-bearing upper structure **85** and/or its upper surface **115** may include first and second surface zones **190** and **195** that may be positioned at least partly over the respective first and second trusses **90** and **95** of the understructure **80**. The first and second surface zones **190** and **195** may, but need not, have identical dimensions, such as identical lengths and widths. The first and second surface zones **190** and **195** may comprise continuous flat surfaces that form part of the upper surface **115**, or they may include features such as longitudinal grooves. Additionally, the first and second surface zones **190** and **195** may be coplanar or they may slope downward from their respective longitudinal edges **200** and **205** toward the midline **110**. Also, the first and second surface zones **190** and **195** may be separated by a bridge **210** that spans at least a portion of the distance between the first and second trusses **90** and **95** of the understructure **80**. The bridge **210** may also be coplanar with the first and second surface zones **190** and **195** or may include a stiffener or other feature such as a flute or medial furrow **215**. The load-bearing upper structure **85** may be bilaterally symmetrical.

(90) The understructure **80** may also be, but need not be, bilaterally symmetrical. The components of the first and second trusses **90** of the understructure **80** may form multiple U-bends. For example, in the embodiment depicted in FIG. 5, the first strut **100**, the first cross beam **105**, and the second strut **125** may form a U-bend; the third strut **140**, the third cross beam **145**, and the fourth strut **150** may form a U-bend; the fifth strut **155**, the fourth cross beam **160**, and the sixth strut **165** may form a U-bend; and the seventh strut **175**, the sixth cross beam **180**, and the eighth strut **185** may form a U-bend. These U-bends may have similar or different sizes. Additionally, the second strut **125**, the second cross beam **130**, and the third strut **140** may form an inverted U-bend; and the sixth strut **165**, the fifth cross beam **170**, and the seventh strut **175** may form an inverted U-bend. The inverted U-bends may have similar or different sizes with respect to each other or with respect to the non-inverted U-bends. One will appreciate that while the embodiment shown in FIG. 5 exhibits rounded edges (such as the edges of the U-bends or the edges of the load-bearing upper structure) that may facilitate rolling or extrusions process, these edges may be manufactured to form substantially transverse angles that may, or may not, be perpendicular.

(91) One or more of the U-bends, the central space **220** under the bridge **210**, or other features of the substructure **80** may form longitudinal channels to provide passage for a hydraulic line (not shown) that conveys hydraulic power from a hydraulic power source (not shown).

(92) FIGS. 6A and 6B (collectively FIG. 6) illustrate a front left isometric view and a cross-sectional view, respectively, of an elongate body **50a**, according to an alternative embodiment; and FIG. 6C illustrates a front left isometric view of an elongate body **50b**, according to another alternative embodiment. With reference to FIG. 6, the functions of the elongate body **50a** and the elongate body **50b** (collectively elongate body **50**) are the same; however, their cross-sectional profiles are different. The elongate body **50a** includes a load-bearing upper structure **240** extending longitudinally from a first end **245** to a second end **250**. The load-bearing upper structure **240** comprises one or more surface zones **255**, such as flat surfaces, and is coupled to a substructure **260** that may include a first truss **265** and a second truss **270** as discussed below. In some embodiments, the surface zone(s) **255** may be rigidly connected to one or more stiffeners **275**, such as flutes. Stiffeners may be integrally formed with the surface zone(s) **255** to accomplish a rigid

connection or may be welded or otherwise suitably secured to the flat surfaces. Stiffeners **275** may provide resistance against longitudinal bending of the flat surfaces, such as the surface zones **255**. Alternate stiffeners include inverted flutes, fins **350** (see, e.g., FIG. 7), and other suitable structures that inhibit longitudinal bending of the flat surfaces. Stiffeners may protrude above the flat surfaces or may protrude below the flat surfaces.

(93) The substructure **260** forms a structural element that resists one or more of flex, torsion, axial compression, and/or lateral deflection of the load-bearing upper structure **240**. The first truss **265** includes a first strut **280** that extends downward from the outer edge of the load-bearing upper structure **240** in a generally orthogonal orientation with respect to the load-bearing upper structure **240**. A first cross beam **285** is coupled to the first strut **280** and extends away from the first strut **280**, for example, substantially orthogonally from the first strut **280** (toward the midline **290** of the elongate body **50**) to form a lower surface **295** of the elongate body **50**. A second strut **300** is coupled to the first cross beam **285** and extends from the first cross beam **285** towards the load-bearing upper structure **240**. The second strut **300** may be non-perpendicular (e.g., positioned in a diagonal plane) with respect to the load-bearing upper structure **240** to enhance the stiffness and torsion-resistance of the elongate body **50**.

(94) A second cross beam **305** is coupled to the second strut **300** and may contact a lower surface **310** of the load-bearing upper structure **240**. The second cross beam **305** can be coupled to the load-bearing upper structure **240**, for example, via spot welds or by being integrally formed with the load-bearing upper structure **240**. A third strut **315** is coupled to the second cross beam **305** and extends from the second cross beam **305** away from the load-bearing upper structure **240**. The third strut **315** may be non-perpendicular (e.g., positioned in a diagonal plane) with respect to the load-bearing upper structure **240** to enhance the stiffness and torsion-resistance of the elongate body **50**. A third cross beam **320** is coupled to the third strut **315** and extends away from the third strut **315** (away from the midline **290** of the elongate body **50**) to form another lower surface **295** of the elongate body **50**. A fourth strut **325** extends from the third cross beam **320** towards the load-bearing upper structure **240** and is coupled to the other outer edge of the load-bearing upper structure **240**.

(95) In some embodiments, the substructure **260** may be coupled to the load-bearing upper structure **240** via welding. The substructure **260** may be integrally formed with the load-bearing upper structure **240**. Alternatively, the substructure **260** may be partially integrally formed with the load-bearing upper structure **240** and secured to the load-bearing upper structure **240** via welding or other suitable attachment. Likewise, elements of the substructure **260** may be integrally formed together, may be welded or otherwise suitably attached together, or may be coupled via a combination of integral formation and attachment such as welding.

(96) The second cross beam **305** may be substantially parallel to the load-bearing upper structure **240**, may contact and/or be secured to or formed as part of the lower surface **310** of the load-bearing upper structure **240**, and may act as a second load-bearing member. Coupling the second cross beam **305** and the load-bearing upper structure **240** to one another, or forming them together, may reduce sliding between them when they are placed under load.

(97) The load-bearing upper structure **240** and the first strut **280**, the first cross beam **285**, and the second strut **300** of the first truss **265** may form a side channel **330**. The load-bearing upper structure **240** and the third strut **315**, the third cross beam **320**, and the fourth strut **325** of the second truss **270** may form a side channel **335**. The first truss **265** and the second truss **270** are spaced apart to form a central channel **340** within the substructure **260** of the fork assembly **40**. These channels may provide passage for a hydraulic line (not shown) that conveys hydraulic power from a hydraulic power source (not shown).

(98) With reference to FIG. 6C, the functions of the elongate body **50b** and the elongate body **50a** (collectively elongate body **50**) are the same; however, their cross-sectional profiles are different. The elongate body **50b** of FIG. 6C is a squatter variation of the elongate body **50a** shown in FIG. 6A. Accordingly, FIGS. 6A and 6C share many of the same reference numerals.

(99) FIG. 7 illustrates a cross-sectional view of another alternative elongate body 50c. With reference to FIG. 7, the elongate body 50c and the elongate body 50 (collectively elongate body 50) are substantially identical with the exception of the stiffener fins 350. The plurality of the fins 350 may serve to form the upper-most surface of the load-bearing upper structure 85, and may support a load, such as a pallet, thereupon.

(100) FIGS. 8A, 8B, and 8C (collectively FIG. 8) illustrate a front right bottom isometric view, a top front right isometric view, and a top front right enlarged isometric view of portions of the elongate body 50 of the modular fork assembly 40, wherein the elongate body 50 is attached to an embodiment of endcaps 360 that form part of a first interlocking mechanism 365 (FIG. 4) that is configured for detachable connection of the elongate body 50 to the load wheel module 55. With reference to FIGS. 3, 4, and 8, the endcaps 360 may be configured for welding or other means of permanent or nonpermanent attachment to the proximal and/or distal ends 42 and 70 of the elongate body 50. In one example, the endcap 360 may have an endcap body 385 (FIG. 4D) between an external side 370 and an internal side 375 (FIG. 4E) with respect to how the endcap 360 is positioned on the end of the elongate body 50, such that the internal side 375 is closest to the elongate body 50 and the external side 370 is farthest from the elongate body 50 and exposed outwardly when the endcap 360 is connected to the elongate body 50.

(101) The endcap 360 may include a beveled edge 390 between a side surface 395 and the internal side 375. The beveled edge 390 may facilitate a snug fit between the endcap 360 and the proximal end 42 and/or distal end 70 of the elongate body 50. The beveled edge 390 may also provide adequate space or surface area to accommodate a superior weld between the endcap 360 and the elongate body 50.

(102) The endcap 360 may include one or more external flanges 380 that extend from the internal side 375 and are configured to slide into one or more of the externally open channels formed in the understructure 80 and/or in the load-bearing upper structure 85. When welding is the desired means of permanent attachment, the flanges 380 may be configured to contact, and/or conform to the shape of, one or more exposed surfaces of the externally open channels, such as the exposed surfaces of the cross beams 130 and 170 or the lower surface 135 beneath the bridge 210.

(103) The endcap 360 may also include one or more prongs (not shown) that extend from the internal side 375 and are configured to slide into one or more of the closed channels formed by the understructure 80 and/or formed by a combination of portions of the understructure 80 and the load-bearing upper structure 85, such as closed channels 400. The prongs may be configured to contact, and/or generally conform to the shape of, one or more of the surfaces of the closed channels 400, such as the interior surfaces of the struts 100, 125, 140, 165, and 175, such as the interior or exterior surfaces of struts 150 and 155, and/or such as the lower surface 135 within the channels 400. Each prong may include one or more tabs or bumps that are configured to mate with crimps or indentations within the interior surfaces of the channels 400 and may serve to hold or secure the endcaps 360 in place while they are welded or more permanently attached by other means to the elongate bodies 50.

(104) The endcaps 360 may form part of the first interlocking mechanism and may include interlocking features that mate with interlocking features of alternative endcaps (not shown) or interlocking features configured into the body-facing end 65 of the load wheel module 55. One example of interlocking features includes one or more sheer-resistant features, such as protruding features 415 and respective mating receiving features 420 (FIG. 4F). In particular, one of the protruding feature 415 or the receiving feature 420 may be configured to permanently connect to the elongate body (or the endcap 360), and a different one of the protruding feature 415 or the receiving feature 420 may be configured to permanently connect to the load wheel module 55 (or an alternative endcap if one is employed for the load wheel module 55).

(105) In one example, the protruding features 415 may protrude from an external face 425 of the external side 370 of the endcap 360 and may have any shape that is adequate for engagement with

respective receiving features **420** configured in alternative endcaps or the interlocking features in the body-facing end **65** of the load wheel module **55**. In particular, the protruding features **415** may have a sectional profile of any geometric shape. For example, the sectional profile of the protruding features **415** may be circular as shown in FIG. **8**, or the sectional profile may be another shape such as elliptical, rectangular, square, triangular, hexagonal, or octagonal.

(106) The protruding features **415** may have any width or diameter that is typically smaller than dimensions of the external face **425** of the external side **370** (such that the protruding features **415** does not extend above the load-bearing upper structure **85**). A typical minimum width of a protruding feature **415** might be about 25 mm. The protruding features **415** may have any protruding height beyond the external face **425** of the external side **370** of the endcap **360** to the extent that the mated receiving feature **420** would not interfere with operation of the load wheel module **55**. A typical minimum protruding height might be about 8 mm.

(107) Similarly, the receiving features **420** may have a sectional profile of any geometric shape. For example, the sectional profile of the receiving features **420** may be circular as shown in FIG. **4F**, or the sectional profile may be elliptical, rectangular, square, triangular, hexagonal, or octagonal. Moreover, the receiving features **420** may have any width or diameter that is typically smaller than dimensions of the external face **425** of the external side **370**. A typical minimum width of a receiving feature **420** might be about 25 mm. The receiving features **420** may have any depth into the mated interlocking feature to the extent that the mated receiving feature **420** would not interfere with operation of the load wheel module **55**. A typical minimum depth of receiving feature **420** might be about 8 mm.

(108) Although FIG. **8** depicts only two mated sets of protruding and receiving features **415** and **420**, additional sets, or only one, may be employed. For example, a set could additionally or alternatively be employed in proximity to one or more of the corners **435** of the external face **425** (and respective face **450**) of the body-facing end **65** of the load wheel module **55**. One will also appreciate that the endcap **360** may employ one or more of the receiving features **420** instead of the protruding features **415**. Alternatively, the endcap **360** may employ a mix of one or more of the receiving features **420** and one or more of the protruding features **415**. The mated endcaps or the interlocking features configured into the body-facing end **65** of the load wheel module **55** would be configured accordingly to mate with the alternative configurations.

(109) With reference again to FIG. **4A**, the first interlocking mechanism **365** may additionally or alternatively employ one or more fastener sets, such as male and female mated fasteners. Any suitable mated fastener set can be employed, such as threaded fasteners, unthreaded fasteners, or compression fasteners. The embodiment shown in FIG. **4** employs externally threaded fasteners **460** that are mated to one or more respective internally threaded receptacles **465**. The externally threaded fasteners **460** may include a head **470**, a shank **475**, a thread **480**, and a tip **485**. The internally threaded receptacles **465** may include a head side **490**, which is closest to the head **470** when the externally threaded fastener **460** is mated to the internally threaded receptacle **465**, and a tip side **495**, which is closest to the tip **485** when the externally threaded fastener **460** is mated to the internally threaded receptacle **465**.

(110) An example of externally threaded fasteners **460** includes 16-mm flange-head cap screws or bolts. One will appreciate that the diameter can be of any suitable size. However, a diameter large enough to aid in resistance to sheer between the detachably connected parts might be beneficial. In some embodiments, the externally threaded receptacles **465** have a minimum shank diameter of about 12 mm.

(111) In the depicted embodiment, the endcap **360** includes multiple ones of the same set of fasteners in the same respective engagement arrangements, such that all of the internally threaded receptacles **465** are configured with their head sides **490** closest to the external face **425** of the endcap **360**. However, one will appreciate that the endcap **360** could include one or more internally threaded receptacles **465** configured with their head sides **490** closest to the external face **425** and

one or more internally threaded receptacles **465** configured with their tip sides **495** (hidden within the flange **480** in connection with the interlocking mechanism **360** but shown in FIG. **3D** in connection with a second interlocking mechanism **675**) closest to the face **425**. Alternatively or additionally, the endcap **360** may be configured to include one or more different types of sets of mated fasteners. In the embodiment shown in FIG. **4E**, the tip side **495** is entirely hidden within the flange **480**, but the internally threaded receptacles **465** could be configured so that they form a tunnel all the way through the flanges **480**.

(112) In the example shown in FIG. **4**, the sheer-resistant features are configured to incorporate one internally threaded receptacles **465**. In particular, the protruding features **415** and the receiving features **420** each have holes to receive the externally threaded fastener **460**. When threaded fasteners are employed, at least the protruding feature **415** or the receiving feature **420** that is intended to be closest to the tip **485** may be internally threaded (if a separate nut, for example, is not employed). In the embodiment shown, the protruding features **415** of the endcap **360** include the internally threaded receptacles **465** because the welding flanges **380** could be in the way of placing a nut or tightening the fastener head **470**. One will appreciate that the design of the elongate body **50** or the endcap **360** can be modified to accommodate placement of the head **470** of the externally threaded fastener **460**, in which circumstances the sheer-resistant feature, such as the receiving feature **420** in the body-facing end **65** of the load wheel module **55** could be configured to incorporate the internally threaded receptacles **465**. However, if the fastener head **470** is to be closest to the body-facing end **65** of the load wheel module **55**, then the shank holes **500** of the receiving feature **420** in the body-facing end **65** of the load wheel module **55** can be threaded or unthreaded.

(113) FIG. **9A** illustrates a front right bottom isometric view of a load wheel module **55** with a load wheel unit **510** in an undeployed position **440**; FIG. **9B** illustrates a front right bottom isometric view of the load wheel module **55** with a load wheel unit **510** in a deployed position **445**; and FIG. **9C** illustrates a front right top isometric view of a wheel module substructure **515**. With reference to FIGS. **9A**, **9B**, and **9C** (collectively FIG. **9**), and additionally FIGS. **3** and **4**, the load wheel module **55** includes a frame **520** that houses the wheel module substructure **515**. The frame **520** includes a frame upper surface **525** and a frame lower surface **530**. The frame upper surface **525** may be configured to support and provide sliding contact to a load, and the frame lower surface **530** may be configured to provide one or more points of contact with components of the wheel module substructure **515**.

(114) The frame **520** may have a body-facing end **65** and a fork tip-facing end **75** that may be substantially identical in shape or that may be different. For example, both of the body-facing end **65** and the fork tip-facing end **75** may be configured to include substantially identical interlocking mechanism components. In particular, the sheer-resistant features, such as the protruding features **415** or the receiving features **420** may be identically positioned on both of the body-facing end **65** and the fork tip-facing end **75**. The wheel module assembly **55** depicted in FIG. **9** includes receiving features **420a** configured into its fork tip-facing end **75**. One will appreciate, however, that the body-facing end **65** and the fork tip-facing end **75** may have different types of sheer-resistant features in the same or different locations on each of the facing ends of the frame **520**.

(115) The frame **520** may also include an aperture **535** in both of the body-facing end **65** and the fork tip-facing end **75** if symmetry is desired for manufacturing. However, in some embodiments, only the body-facing end **65** of the frame **520** may include the aperture **535** to provide passage for a hydraulic line (not shown) that conveys hydraulic power from a hydraulic power source (not shown) to a hydraulic actuator **540** in the load wheel module **55**. In other embodiments, the aperture **535** may provide passage for a mechanical link arm to facilitate lowering and raising a load wheel **565** via a suitable mechanical system coupled to the load wheel module **55**, where the mechanical link arm receives a motive force from a power source located in a forklift truck body.

(116) The wheel module substructure **515** may include a hydraulic actuator assembly **545** and a

load wheel assembly **550** that is operatively connected to the frame **520**. The load wheel assembly **550** includes a wheel carrier strut **555** (also called a wheel carrier frame) that is operatively connected to, and supports, the load wheel unit **510** that includes a wheel carrier **560** that supports one or more load wheels **565**. In one example, the wheel carrier strut **555** has a U-shaped distal portion that is pivotally connected to the wheel carrier **560** on both sides of the load wheels **560**. (117) The wheel module substructure **515** may be positioned within the frame **520** such that the hydraulic actuator assembly **545** is positioned closer to the body-facing end **65** of the load wheel module **55** and the load wheel assembly **550** is positioned closer to the fork tip-facing end **75** of the load wheel module **55**. In particular, the hydraulic actuator **540** may be positioned closer to the body-facing end **65** and the load wheels **565** may be positioned closer to the fork tip-facing end **75**. (118) The wheel carrier strut **555** is also operatively connected to the load wheel module frame **520** and to the hydraulic actuator assembly **545**. In one example, the operative connection to the frame **520** may be implemented by one or more pivot bars **570** that may be pivotally connected at a bar frame end **575** to the frame **520** and at a bar strut end **580** to the wheel carrier strut **555**. Part of a pivot mechanism **585** at the bar frame end **575** may be secured within a recess **590** in an exterior side surface **595** of the frame **520** so that the part of the pivot mechanism **585** will not catch when the modular fork assemblies **40** are slid into load structures that support the load. One will appreciate that other pivot mechanisms can additionally or alternatively be counter sunk into the components that they are pivoting. For example, although not depicted in this manner, part of the pivot mechanism **600** at the bar strut end **580** may be recessed into the pivot bar **570**. (119) An actuator-facing end **605** of the wheel carrier strut **555** may be operatively connected to the hydraulic actuator assembly **545** via a pivot mechanism **610** at a strut-facing end **615** of the hydraulic actuator assembly **545**. The pivot mechanism **610** may include a pivot **620** that extends through one or more strut teeth **625** at the actuator-facing end **605** of the wheel carrier strut **555** that are interweaved with one or more actuator assembly teeth **630** at the strut-facing end of the hydraulic actuator assembly **545**. (120) The hydraulic actuator assembly **545** may include a hydraulic line input connector (also called a cap-end port) (not shown) operative for connecting the hydraulic actuator **540** to a hydraulic line (not shown) that transmits hydraulic fluid from a hydraulic power source (not shown). The hydraulic line input connector may supply a hydraulic manifold **640** that distributes hydraulic power from the hydraulic line into multiple hydraulic barrels (also called hydraulic cylinders) **645** that each include a piston **650** (shown in broken lines in FIG. 9C) that is operatively connected to a piston rod **655**. (121) In some embodiments, the hydraulic actuator **540** may include from one to ten pistons **650**. In some embodiments, the hydraulic actuator **540** includes at least two pistons **650**. In some embodiments, the hydraulic actuator **540** may include from two to ten pistons **650**. In some embodiments, the hydraulic actuator **540** includes at least three pistons **650**. In some embodiments, the hydraulic actuator **540** may include from three to eight pistons **650**. In some embodiments, the hydraulic actuator **540** may include from three to six pistons **650**. FIG. 9C shows an example of a hydraulic actuator **540** that includes four hydraulic barrels **645**, each of which includes a respective piston **650**. (122) In one example, the piston **650** may have has a length within the range of about 0.50 inches to about 3 inches (or about 1.25 cm to about 7.75 cm) and a stroke length within the range of about 1 inch to about 3 inches (or about 2.50 cm to about 7.75 cm). In another example, the piston **650** may have a length within the range of about 1 inch to about 2 inches (or about 2.50 cm to about 5.25 cm) and a stroke length within the range of about 1.5 inches to about 2.5 inches (or about 3.75 cm to about 6.50 cm). (123) In one example, the hydraulic barrels **645** have a capability up to about 3,200 psi. In some embodiments, the hydraulic barrels **645** have a capability of greater than about 2,000 psi. In some embodiments, the hydraulic barrels **645** have a capability of greater than about 3,000 psi. In some

embodiments, the hydraulic power from the hydraulic power source has a maximum pressure within the range of about 13,790 to about 27,580 kilopascals (about 2000 to about 4000 psi) at the hydraulic line input connector.

(124) In some embodiments, the hydraulic actuator **540** is operable to provide maximum thrust within a range of about 15,000 pounds (or about 66,700 newtons) to about 30,000 pounds (or about 133,500 newtons). In some embodiments, the hydraulic actuator **540** is operable to provide greater than about 15,000 pounds (or greater than about 66,750 newtons) of thrust. In some embodiments, the hydraulic actuator **540** is operable to provide greater than about 20,000 pounds (or greater than about 89,000 newtons) of thrust.

(125) The load wheel unit **510** may rest in an undeployed position **440** when the hydraulic actuator **540** is not actively pushing the piston rods **655** beyond a cylinder head **660** of the piston assembly. The load wheel unit **510** may be deployed into a deployed position **445** in response to a load wheel deployment signal that may be provided by an automated system or may be provided in response to a manually activated input, such as a switch or button. The load wheel deployment signal directly or indirectly causes hydraulic power to be propagated through a hydraulic line positioned within the elongate body **50** of the modular fork assembly **40**. The hydraulic power may be in the form of a hydraulic fluid under pressure.

(126) The hydraulic line delivers the hydraulic power through the hydraulic line input connector to the hydraulic manifold **640** that distributes the hydraulic power to the hydraulic barrels **645** of the hydraulic actuator **540**. The hydraulic power pushes the pistons **650** of the hydraulic actuator **540** so that the piston rods **655** extend beyond the cylinder head **660** to push against the actuator-facing end **605** of the wheel carrier strut **555**, causing the pivot bar **570** to force the load wheel unit **510** to assume a predetermined deployed position **445** in which the load wheel unit **510** is vertically spaced apart from the load wheel module frame **520**. One will appreciate that the hydraulic line and hydraulic actuator assembly **545** can be replaced by a link rod that is actuated close to the proximal end **40** of the elongate body **50** and a mechanical system coupled to the load wheel module **55** and arranged to lower and raise the load wheel **565** in response to movement of the link rod. For example, a suitable mechanical system may be coupled to a load wheel module **55** with a link rod extending through an elongate body **50** of a fork assembly **40** to mechanically connect the mechanical system with a power source such that force from the power source is transmitted via the link rod to the mechanical system to lower and raise the load wheel **565**.

(127) With reference again to FIGS. **3** and **4**, a second interlocking mechanism **675** can be configured to detachably connect the load wheel module **55** to the fork tip **60** or an endcap **670**. The fork-tip facing end **75** of the load wheel module **55** can be configured to mate with the proximal end **665** of the fork tip **60** or the endcap **670**. The endcap **670** may form part of the second interlocking mechanism **675**. The endcap **670** may be substantially identical to, or different from, the endcap **360**. Moreover, the second interlocking mechanism **675** may be substantially identical to, or different from, the first interlocking mechanism **365**. In the embodiment shown in FIG. **3D**, the internally threaded receptacle **465** may be configured as a ferrule **685** that protrudes from the internal side of the endcap **670**. The tip side **495** of the internally threaded receptacle **465** may be open or closed.

(128) All the alternatives described with respect to the first interlocking mechanism **365** may apply to the second interlocking mechanism **675**. In one alternative embodiment, the fork-tip facing end **75** of the load wheel module **55** may be provided with the sheer-resistant protruding features **415**, and the endcap **670** or the proximal end **665** of the fork tip **60** may be provided with the sheer-resistant receiving features **420**.

(129) The fork tip **60** has a distal end (also referred to as the toe end) **680** that is opposite the proximal end **665**, i.e., the distal end **680** is furthest from the battery box **41**. The distal end **680** of the fork tip **60** initially engages a pallet when the modular fork assembly **40** is directed to pick up a load. The fork tip **60** may taper in one or more dimensions from the proximal end **665** to the distal

end **680** so that the perimeter of the distal end **680** is smaller than the perimeter of the proximal end **665**. In one example, the fork tip **60** and/or the distal end **680** has a wedged shape. In another example, the distal end **680** has a curved shape.

(130) A major advantage of the modularity of the modular fork assembly **40** is that any one of the modular components, such as the elongate body **50**, the load wheel module **55**, or the fork tip **60**, can be readily replaced if they become bent or otherwise damaged. Such replacement can be achieved without metal cutting or welding. In some embodiments, only a simple tool such as a screwdriver or wrench may be utilized to effect the replacement. Moreover, these modular fork assemblies **40** and their components may be readily salvaged from a disabled vehicle and reused in a working vehicle or as replacement parts.

(131) One approach that increases the success ratio of pallet engagement and disengagement involves offsetting the forward load wheels **14** of the different forks **12** along a longitudinal axis **18** of the pallet truck **9**. This approach is disclosed in detail in U.S. Provisional Patent Application No. 62/609,235, entitled "Offset Load Rollers for a Pallet Truck", which is incorporated herein by reference.

(132) Load wheels with torque-coupling assemblies may be used in conventional fork configurations or other suitable environments as well. This disclosure teaches additional or alternative means to increase the entry and exit potential with respect to pallet pockets **6** of closed pallets **1** that are empty or have a light-weight palletized load. For example, torque between the rear load wheel **15** and the forward load wheel **14** can be coupled so that if either of the rear load wheel **15** or the forward load wheel **14** meet resistance at a base board **2**, then rotation of the other of the load wheels **17** will increase the ability of the inhibited load wheel **17** to move over the base board **2**. In particular, if the forward load wheel **14** meets resistance when encountering the base board **2** while attempting to enter a pocket **6**, then the continuous rotation of the rear load wheel **15** in a forward direction (caused by friction against the ground due to movement of the pallet truck **9**) will cause rotation of the forward load wheel **14** if the torque is coupled between the load wheels **17**. Similarly, if the rear load wheel **15** meets resistance when encountering a base board **2** while attempting to exit a pocket **6**, then the continuous rotation of the forward load wheel **14** (in a reverse direction) will cause rotation of the rear load wheel **15** if the torque is coupled between the load wheels **17**.

(133) FIG. **10A** is a top plan view of a load wheel assembly **21** employing an embodiment of a torque-coupling assembly **78**, and FIG. **10B** is a side elevation view of the load wheel assembly **21** of FIG. **10A** with one of two wheel carrier struts **22** and a portion of a wheel-carrier bracket **86** removed for clarity. A torque coupler or torque-coupling assembly **78** can be implemented in many ways. One will appreciate that a torque-coupling assembly **78** can be implemented as any torque coupler known in the art, as well as any of the torque-coupling means for coupling torque described herein.

(134) With reference to FIGS. **10A** and **10B**, the load wheel assembly **21** may include one or more wheel-carrier brackets **86** that operatively connect one or more of the load wheels **17** to the fork **12**. The wheel-carrier bracket **86** may include one or more fork attachment pivot holes **23** or **23a** that can be used to attach the wheel-carrier bracket **86** to the fork **12**. The fork attachment pivot holes **23** may be positioned toward a proximal end of the wheel-carrier bracket **86** on separate bracket arms **24** connected by a bracket cross bar **25**. The wheel-carrier struts **22** may extend forward from the bracket cross bar **25** on separate sides of the load wheels **17**. In some embodiments, each wheel-carrier strut **22** may include an axle mount hole **26** (or other mounting means) adapted to receive a carrier axle **27** (with or without bushings **95**, see FIG. **1E**) of a wheel carrier **29** to facilitate rotatable connection between the wheel-carrier struts **22** and the wheel carriers **29**. The wheel carriers **29** are also connected to a forward axle **30** and a rear axle **31** to permit rotation of the respective forward load wheel **14** and the rear load wheel **15**. For purposes of discussion, the wheel carrier **29**, the forward load wheel **14**, the rear load wheel **15**, the forward axle **30**, the rear

axle **31**, and the torque-coupling assembly **78** can be grouped together as a load wheel unit **78**.

(135) Forks **12** tend to have a limited height dimension that is typically shorter than a spacer height **7** of the closed-bottom pallet **1** so that the forks **12** and their respective load wheel assemblies **21** can easily fit between the base boards **2** and the upper planks **5**. The spacer height **7** may be slightly shorter than a pallet gap height **8** between the base boards **2** and the upper planks **5**. Accordingly, the forward load wheel **14** has a forward wheel radius **102** that may be less than half the spacer height **7** or less than half the pallet gap height **8**. Similarly, the rear load wheel **15** has a rear wheel radius **104** that may be less than half the spacer height **7** or less than half the pallet gap height **8**. Additionally, the forward load wheel **14** has a forward wheel height **106** above the supporting surface **3** that may be shorter than the spacer height **7** or the pallet gap height **8**. Similarly, the rear load wheel **15** has a rear wheel height **108** above the supporting surface **3** that may be shorter than the spacer height **7** or the pallet gap height **8**.

(136) One will appreciate that the operative connection of the load wheels **17** to the fork can be implemented in a variety of alternative ways. For example, FIGS. **17A-17C** and **17A-17C** show one embodiment for connecting the load wheels **17** to the fork and are described later.

(137) Neither the forward load wheel **14** nor the rear load wheel **15** is connected directly or indirectly to a motor, i.e., the load wheels **17** are non-motorized or unmotorized. More specifically, there is no mechanical linkage from a drive motor that causes rotation of the load wheels **17**. Moreover, rotation of the load wheels **17** is caused by movement of the pallet truck **9** and contact of at least the forward load wheel **14** or the rear load wheel **15** with the supporting surface **3**.

(138) The torque-coupling assembly **78** shown in FIGS. **10A** and **10B** employs an idler wheel **116** (also called a coupling wheel) that may rotate about an idler axle **121** and that has an idler wheel surface **76** (also called a coupling wheel surface) that directly contacts both the forward load wheel **14** and the rear load wheel **15**. The idler wheel **116** may utilize a tire that can have a pneumatic or a solid core, such as used in any conventional tire. The surface **76** of the idler wheel **116** and/or the core of the idler wheel **116** can be made from the same or different materials. In some embodiments, the surface **76** and/or the core of the idler wheel **116** can be made from a polymer, such as polyurethane. In other embodiments, the surface **76** and/or the core of the idler wheel **116** can be made from a metal, such as aluminum or steel. In some embodiments, the surface **76** of the idler wheel **116** may employ a non-smooth texture.

(139) The idler wheel **116** may have a small idler wheel diameter **114** so as to provide some clearance with an upper wall interior surface of the fork **12**. In particular, the idler wheel diameter **114** may be shorter than or equal to the forward wheel radius **102** and/or the rear wheel radius **104**. Moreover, the idler wheel **116** has an idler wheel height **118** above the supporting surface **3** that may be higher than or equal to the forward wheel height **106** or the rear wheel height **108**, or the idler wheel height **118** may be shorter than or equal to the forward wheel height **106** or the rear wheel height **108**. In some embodiments, the idler wheel height **118** may be within 5% of the forward wheel height **106** or the rear wheel height **108**.

(140) The amount of torque transferred by the torque-coupling assembly **78** is highly variable depending on materials used, surface finishes, and contamination from the environment. A suitable range for torque transfer may be from about 1 to 40 Newton meters (Nm) or may be from about 2 to 20 Newton meters (Nm). Generally, the amount of torque transferred by the torque-coupling assembly **78** is greater than or equal to 2 Nm. One will appreciate, however, that the amount torque transferred may be less than 2 Nm. One will also appreciate that the amount torque transferred may be greater than 40 Nm.

(141) This torque-coupling assembly **78** also employs one or more or force-applying coupling assemblies or force-applying couplers **122**. A force-applying coupler or force-applying coupling assembly **122** can be implemented in many ways. One will appreciate that a force-applying coupling assembly **122** can be implemented as any force-applying coupling assembly known in the art, as well as any of the force-applying coupling means for coupling force described herein. In

many embodiments, such as any of the force-applying coupling assemblies **122** (with or without an additional letter designation), the force-applying coupling assembly **122** may fall into the subcategory of a resilient force-applying coupling assembly (or resilient force-applying coupler). (142) The force-applying coupler **122** shown in FIGS. **10A** and **10B** also constitutes a resilient force-applying coupler **122**, employing a torsion spring **124**. The torsion spring **124** can have a single loop **126** or can have a helical component. The torsion spring **124** may include an idler arm **128** that is directly or indirectly connected to the idler wheel **116** such as to the idler axle **121**. The torsion spring **124** may also include a carrier arm **131** that is directly or indirectly connected to the wheel carrier **29**, such to its carrier axle **27**. In some embodiments, the idler arms **128** may have auxiliary projections **188** that insert into the idler axle **121** of the idler wheel **116** or that function as the axle **121** of the idler wheel **116**. In some embodiments, the idler arms **128** may slide through an axle slot **138** in an idler axle **121b** of an idler wheel **116b** as shown in FIG. **12C**. The slot **138** may prevent rotation of the idler axle **121** in embodiments in which the idler wheel **116** constitutes a roller with bearings over the idler axle **121**.

(143) In the embodiment shown in FIGS. **10A** and **10B**, the loop **126** of the torsion spring **124** is positioned around the rear axle **31** and the idler arm **128** crosses carrier arm **131** to provide force (such as tension) to hold the idler wheel **116** against the forward load wheel **14** and the rear load wheel **15**. One will appreciate however that the loop **126** of the torsion spring **124** could be positioned about the forward axle **30** instead of the rear axle **31**. Or, the loops of two torsion springs **124** could be positioned about both the forward axle **30** and the rear axle **31** to provide additional force (and/or a backup coupling) if desired. If torsion springs **124** are positioned on both sides (left and right) of the load wheels **14** and **15**, then the loops **126** of both torsion springs **124** may be positioned about the same axle. In some embodiments, however, the loops **126** may be positioned about different axles. One will also appreciate that the force-applying coupling assembly **122**, and more specifically the loops **126** of the torsion springs **124**, can alternatively be positioned outside the wheel carrier **29**, i.e. between the wheel carrier **29** and the strut **22**, on an extension of the forward axle **30** or the rear axle **31**. Such an axle extension could include a larger diameter rim to prevent the loop **126** from coming off the axle extension.

(144) One advantage of employing a force-applying coupler **122** is that it can be adapted to apply enough force to couple the torque between the forward load wheel **14** and the rear load wheel **15** (e.g., ensure that there is sufficient friction between the forward load wheel **14** and the rear load wheel **15**), and at the same time not create unnecessary drag on the movement of the forward load wheel **14** and the rear load wheel **15**. Suitable force provided by the cumulative force-applying couplers **122** acting on the forward load wheel **14** and the rear load wheel **15** may be in the range of about 5 to 315 Newtons, or the force may be in the range of about 5 to 110 Newtons. Generally, the cumulative coupling force may be greater than or equal to about 5 Newtons. One will appreciate that the cumulative coupling force may be less than 5 Newtons. One will also appreciate that the cumulative coupling force may be greater than 315 Newtons. Additionally, one will appreciate that this cumulative force can be divided by the total number of springs in the total number of force-applying couplers **122** that are employed in any given torque-coupling assembly **78** to determine a desirable amount of force per force-applying coupler **122**.

(145) One will also appreciate that the force coupling (or resilient force coupling) of the idler wheel **116** to the load wheels **17** can be implemented in a variety of other ways. For example, FIG. **11A** is a top plan view of a load wheel assembly **21a** employing an alternative embodiment of a torque-coupling assembly **78a**, and FIG. **11B** is a side elevation view of the load wheel assembly **21a** of FIG. **10A** with one of the wheel-carrier struts **22** and a portion of the wheel-carrier bracket **86** removed for clarity. The alternative torque-coupling assembly **78a** may employ an alternative force-applying coupler (or resilient force applying coupler) **122a** that includes an alternative to the torsion spring **124** in the form of a spring plate or leaf spring **124a**. The leaf spring **124a** may include spring end tabs **111** that extend from a central curved portion **112** of the leaf spring **124a**.

The leaf spring **124a** may be connected to optional bosses or ridges **132** of an alternative wheel carrier **29a** by any suitable attachment means such as carrier bolts (or screws) **134**. The idler wheel **116** may have a smaller idler wheel height **118** adapted to accommodate a downward curve of the leaf spring **124a** that can provide force against the idler axle **121** so that the leaf spring **124a** urges the idler wheel **116** against the forward load wheel **14** and the rear load wheel **15**. FIG. **11C** shows a side elevation view of a leaf spring **124a** of FIG. **11B** in one example of a relaxed shape when the leaf spring **124a** is not yet deployed over the idler axle **121** and connected to the wheel carrier **29**. (146) FIG. **12A** is a top plan view of a load wheel assembly **21b** employing another alternative embodiment of a torque-coupling assembly **78b**, FIG. **12B** is a side elevation view of the load wheel assembly **21b** of FIG. **12A** with a wheel carrier strut **22** and a portion of a wheel carrier bracket **86** removed for clarity, and FIG. **12C** is a side elevation view of an idler wheel **116b** of the torque-coupling assembly **78b**. The alternative torque-coupling assembly **78b** may employ an alternative force-applying coupler (or resilient force-applying coupler) **122b** that includes an alternative spring **124b** in the form of an inverted leaf spring or spring plate. The spring **124b** may be held by spring brackets **136** that form pockets **137** on an alternative wheel carrier **29b** at a position above or near the forward axle **30** and the rear axle **31**, or the spring **124b** may be connected to the wheel carrier **29b** by any suitable attachment means such as carrier bolts (or screws).

(147) The spring **124b** may slide through an axle slot **138** in an idler axle **121b** of the idler wheel **116b** torque-coupling assembly **78b** as shown in FIG. **12C**, to provide force against the idler axle **121b** so that the leaf spring urges the idler wheel **116b** against the forward load wheel **14** and the rear load wheel **15**. As previously noted, the slot **138** may prevent rotation of the idler axle **121b** in embodiments in which the idler wheel **116b** constitutes a roller with bearings over the idler axle **121b**. Alternatively, the spring **124b** may be positioned (not shown) above the idler axle **121b** to provide downward force against the idler axle **121b** so that the leaf spring urges the idler wheel **116b** against the forward load wheel **14** and the rear load wheel **15**. The idler wheel **116b** may have an idler wheel height **118b** adapted to accommodate the curve of the leaf spring (or the curve of the leaf spring may be adapted to idler wheel height **118b**) to provide force against the idler axle **121b** so that the leaf spring urges the idler wheel **116b** against the forward load wheel **14** and the rear load wheel **15**. Although not shown, bolts **134**, screws, pins, or other fasteners can be employed to secure the spring **124b** to the spring brackets **136** from above the spring brackets **136** into holes or slots toward the ends of the springs **124b**.

(148) FIG. **13A** is a top plan view of a load wheel assembly **21c** employing another alternative embodiment of a torque-coupling assembly **78c**, and FIG. **13B** is a side elevation view of the load wheel assembly **21c** of FIG. **13A** with a wheel carrier strut **22** and a portion of a wheel carrier bracket **86** removed for clarity. The alternative torque-coupling assembly **78c** may employ an alternative force-applying coupler (or resilient force-applying coupler) **122c** that includes a spring clip **124c** generally in the form of a “U”-shaped clip, having a lower segment **141**, a bend **142**, and an upper segment **144**. The lower segment **141** of the spring clip **124c** may be a wire or a strip. The lower segment **141** of the spring clip **124c** may be connected to a hole (not shown) on the top **172** of the wheel carrier **29c** at a position above or near the forward axle **30** and the rear axle **31** by any suitable attachment means such as one or more carrier bolts **134** (or screws).

(149) The upper segment **144** of the spring clip **124c** may constitute a pre-loaded spring that is positioned (as shown in FIG. **13A**) above the idler axle **121** to provide downward force against the idler axle **121** so that the spring clip **124c** urges the idler wheel **116** against the forward load wheel **14** and the rear load wheel **15**. Alternatively, the upper segment **144** may slide through an axle slot (not shown, but such as in FIG. **12C**) in an idler axle **121** of the idler wheel **116** of the torque-coupling assembly **78c** to provide force against the idler axle **121** so that the spring clip **124c** urges the idler wheel **116** against the forward load wheel **14** and the rear load wheel **15**. As noted previously, such a slot **138** may prevent rotation of the idler axle **121** in embodiments in which the

idler wheel **116** constitutes a roller with bearings over the idler axle **121**.

(150) The idler wheel **116** may have an idler wheel height **118c** adapted to accommodate the curve and upper segment of the spring clip **124c** (or the curve of the spring clip **124c** may be adapted to idler wheel height **118c**) to provide force against the idler axle **121** so that the spring clip **124c** urges the idler wheel **116** against the forward load wheel **14** and the rear load wheel **15**. One will appreciate that the bolt **134** may be positioned closer to either the forward axle **30** or the rear axle **31** with the bend **142** positioned closer to the opposite axle. When the spring clips **124c** are positioned at both sides of the idler wheel **121**, the bends **142** can be facing the same direction or different directions. FIG. **13C** shows a side elevation view of the spring clip **124c** of FIG. **13B** in one example of a relaxed shape when the spring clip **124c** is not yet deployed about the ends **156** and **158** of the wheel carrier **29**.

(151) FIG. **14A** is a top plan view of a load wheel assembly **21d** employing another alternative embodiment of a torque-coupling assembly **78d**, and FIG. **14B** is a side elevation view of the load wheel assembly **21d** of FIG. **14A** with a wheel carrier strut **22** and a portion of a wheel carrier bracket **86** removed for clarity. The alternative torque-coupling assembly **78d** may employ an alternative force-applying coupler (or resilient force-applying coupler) **122d** that includes a tension clip **124d** generally in the form of a “frog”-shaped clip, having a rounded upper back segment **146**, two upper thigh segments **148**, and a knee bend **151**, two lower leg segments **152**, and two foot segments **154**.

(152) The rounded upper back segment **146** of the tension clip **124d** may be positioned (as shown in FIG. **14A**) above the idler axle **121** to provide downward tension against the idler axle **121** so that the tension clip **124d** urges the idler wheel **116c** against the forward load wheel **14** and the rear load wheel **15**. Alternatively, the rounded upper back segment **146** may slide through an axle slot (not shown, but such as shown in FIG. **12C**) in an idler axle **121** of the idler wheel **116** of the torque-coupling assembly **78d** to provide tension against the idler axle **121** so that the tension clip **124d** urges the idler wheel **116** against the forward load wheel **14** and the rear load wheel **15**. As noted previously, such a slot **138** may prevent rotation of the idler axle **121** in embodiments in which the idler wheel **116** constitutes a roller with bearings over the idler axle **121**. The upper back segment may be flat instead of rounded and may have a length that is longer than the diameter of the idler axle **121** to allow some float of the idler wheel **116** in connection with its contact to the forward load wheel **14** and the rear load wheel **15**.

(153) The upper back segment **146** may adjoin an upper thigh segment **148** on each side that each reach an outward knee bend **151** that may be above and near the forward end **156** and the rear end **158** of the wheel carrier **29**. From the knee bends **151**, lower leg segments **152** project toward the wheel carrier **29** and are attached to foot segments **154** that at least partly wrap around and tensionally engage the ends **156** and **158**. The tension clip **124d** can be slipped onto the wheel carrier **29** without any additional fastener; however, the foot segments **154** or other parts of the tension clip **124d** could be connected to the wheel carrier **29** by bolts or other fasteners. FIG. **14C** shows a side elevation view of the tension clip **124d** of FIG. **14B** in one example of a relaxed shape when the tension clip **124d** is not yet deployed about the ends **156** and **158** of the wheel carrier **29**.

(154) FIG. **15A** is a top plan view of a load wheel assembly **21e** employing another alternative embodiment of a torque-coupling assembly **78e**, and FIG. **15B** is a side elevation view of the load wheel assembly **21e** of FIG. **15A** with a wheel carrier strut **22** and a portion of a wheel carrier bracket **86** removed for clarity. The alternative torque-coupling assembly **78e** may employ an alternative force-applying coupler (or resilient force-applying coupler) **122e** that includes a tension plate **161**, compression springs **162** and shoulder bolts **164**. The tension plate **161** may have a straight upper back segment **166** between two bends **167** for vertical strut segments **168**, which may terminate in plate tabs **171** that may be parallel (as shown) or angled (not shown) with respect to the top of an alternative wheel carrier **29e**. The shoulder bolts **164** may extend through slots or holes (not shown) in the plate tabs **171** and be connected by a threaded portion (not shown) to

threaded holes (not shown) in a surface at the top **172** of the wheel carrier **29e**. The compression springs **162** may be positioned around shanks of the shoulder bolts **164** between bolt heads **202** and the plate tabs **171** to press the plate tabs toward the top **172** of the wheel carrier **29e**. The pressure of the compression springs **162** against the plate tabs **171** causes the tension plate **161** against the idler axle **121**, which causes the idler wheel **116** to press against the forward load wheel **14** and the rear load wheel **15**. One or more of the length of the shoulder bolts **164**, the strength of the compression springs **162**, the diameter **114** of the idler wheel **116**, and the diameter of the idler axle **121** can be adjusted to determine the idler wheel height **118e**.

(155) In an alternative embodiment, the upper back segment **166** may slide through an axle slot (not shown, but such as shown in FIG. **12C**) in an idler axle **121** of the idler wheel **116** of the torque-coupling assembly **78e** to provide tension against the idler axle **121** so that the force-applying coupler **122e** urges the idler wheel **116** against the forward load wheel **14** and the rear load wheel **15**. As noted previously, such a slot **138** may prevent rotation of the idler axle **121** in embodiments in which the idler wheel **116** constitutes a roller with bearings over the idler axle **121**.

(156) FIG. **16A** is a top plan view of a load wheel assembly **21f** employing another alternative embodiment of a torque-coupling assembly **78f**, and FIG. **16B** is a side elevation view of the load wheel assembly **21f** of FIG. **16A** with a wheel carrier strut **22** and a portion of a wheel carrier bracket **86** removed for clarity. The alternative torque-coupling assembly **78f** employs both top and bottom idler wheels **116** that may be identical or different. Differences might include one or more of idler wheel diameter **114**, idler axle diameter, and idler wheel surface textures, and idler wheel compositions, etc. One will appreciate that the torque-coupling assembly **78f** is adapted so that the bottom of the bottom idler wheel **116** is at or above the bottom of the forward load wheel **14** and the rear load wheel **15**. In many embodiments, the idler wheel diameter **114** and/or the spacing between the forward load wheel **14** and the rear load wheel **15** can be adjusted to determine the height of the bottom of the bottom idler wheel **116** from the supporting surface **3**. The spacing might be partly controlled by the strength of an alternative force-applying coupler (or resilient force-applying coupler) **122f**.

(157) The alternative force-applying coupler **122f** of the alternative torque-coupling assembly **78f** may employ a tension spring **124f** that tensions both the top and bottom idler wheels **116** against the upper surfaces **174** and lower surfaces **176** of the load wheels **14** and **15**. The tension spring **124f** may include axle loops **178** that partly or completely circle the idler axles **121** of the top and bottom idler wheels **116**. The tension spring **124f** also includes a spring section **181** that connects the axle loops **178** to tension the idler axles **121** of the top and bottom idler wheels **116** toward each other. This tension urges the top and bottom idler wheels **116** to contact the respective upper surfaces **174** and lower surfaces **176** of the load wheels **14** and **15** and couple the torque of the forward load wheel **14** and the rear load wheel **15**. One or more of the strength of the spring section **181**, the diameter **114** of the idler wheel **116**, and the diameter of the idler axle **121** can be adjusted to determine the relative elevation of the idler wheel surfaces with respect to the surfaces of the forward load wheel **14** and the rear load wheel **15**.

(158) FIG. **17A** is a top plan view of a load wheel assembly **21g** employing another alternative embodiment of a torque-coupling assembly **78g**, and FIG. **17B** is a side elevation view of the load wheel assembly **21g** of FIG. **17A** with a wheel carrier strut **22** and a portion of a wheel carrier bracket **86** removed for clarity. The alternative torque-coupling assembly **78g** may employ a pair of top and bottom idler wheels assemblies **121** that may be identical or different. Differences might include one or more of idler wheel width, idler wheel diameter **114**, idler axle diameter, and idler wheel surface textures, etc.

(159) Each idler wheel assembly **182** includes multiple idler wheels **116a**, such as two idler wheels **116a**, connected by a central axle block **184**. A top idler wheel assembly **182.sub.1** of the alternative torque-coupling assembly **78g** might include idler wheels **116a.sub.1** and **116a.sub.2** having respective idler axles **121a.sub.1** and **121a.sub.2** that are connected by a central axle block

184.sub.1, and a bottom idler wheel assembly **182.sub.2** of the alternative torque-coupling assembly **78g** might include a first bottom idler wheel (not shown) and a second bottom idler wheel **116b.sub.2** having a respective first bottom idler axle (not shown) and a second bottom axle **121a.sub.2** that are connected by a central axle block **184.sub.2**.

(160) An alternative force-applying coupler (or resilient force-applying coupler) **122g** of the alternative torque-coupling assembly **78g** may employ a tension bolt **186** with top and bottom compressions springs **162.sub.1** and **162.sub.2** (collectively compression springs **162**). The tension bolt **186** extends between the forward load wheel **14** and the rear load wheel **15** and through the central axle blocks **184.sub.1** and **184.sub.2**. The compression springs **162** may be positioned around the shank of the tension bolt **186** between bolt heads **202.sub.1** and **202.sub.2** and the respective proximal surfaces of the central axle blocks **184.sub.1** and **184.sub.2** to press them toward each other.

(161) The pressure of the compression springs **162** against the central axle blocks **184.sub.1** urges the idler axles **121a.sub.2** and **121b.sub.2** toward each other (and idler axle **121a.sub.1** toward the other bottom idler axle (not shown)), which causes the idler wheel **116a.sub.2** (and idler wheel **116a.sub.1**) to press against the upper surfaces **174** of the forward load wheel **14** and the rear load wheel **15** and causes the idler wheel **116b.sub.2** (and the other lower idler wheel (not shown)) to press against the lower surfaces **176** of the forward load wheel **14** and the rear load wheel **15**, thereby coupling the torque of the forward load wheel **14** and the rear load wheel **15**. One or more of the strength of the compression springs **162**, the diameter **114** of the idler wheels **116**, and the diameter of the idler axles **121** can be adjusted to determine the relative elevation of the top and bottom idler wheel surfaces with respect to the respective top and bottom surfaces of the forward load wheel **14** and the rear load wheel **15**. One will appreciate that the torque-coupling assembly **78g** can be adapted so that the bottom of the bottom idler wheel **116a.sub.2** is at or above the bottom of the forward load wheel **14** and the rear load wheel **15**.

(162) FIG. **18A** is a top plan view of a load wheel assembly **78h** employing another alternative embodiment of a torque-coupling assembly **78h** with alternative force-applying coupler (or resilient force-applying coupler) **122h** positioned outside the wheel carrier **29**, and FIG. **18B** is a side elevation view of the load wheel assembly **78h** of FIG. **18A** with a wheel carrier strut **22** and a portion of a wheel carrier bracket **86** removed for clarity. FIG. **18C** is a top plan view of a load wheel assembly **78** employing alternative embodiment of a torque-coupling assembly **78h** with a force-applying coupler positioned inside the wheel carrier **29**, and FIG. **18D** is a side elevation view of the load wheel assembly **78h** of FIG. **18C** with a wheel carrier strut **22** and a portion of a wheel carrier bracket **86** removed for clarity. The alternative torque-coupling assembly **78h** employs top and bottom idler wheels **116** that may be identical or different. Differences might include one or more of idler wheel diameter **114**, idler axle diameter, and idler wheel surface textures, etc.

(163) The alternative force-applying coupler **122h** may employ a torsion spring **124h** that forces both the top and bottom idler wheels **116** against the upper surfaces **174** and lower surfaces **176** of the load wheels **14** and **15**. The torsion spring **124h** may include one or more axle loops **126h** (a single loop or a helical component) that circle the one of the forward axle **30** of the forward load wheel **14** or the rear axle **31** of the rear load wheel **15**.

(164) In the embodiment shown in FIGS. **18A** and **18B**, the load wheel axle **30** or **31** or an extension of it, about which the torsion spring **124h** is deployed, may extend out further beyond the wheel carrier **29** to provide greater support for the loop **126h** of the torsion spring **124h**. Such elongated load wheel axle **30** or **31** may extend to, or almost to, the wheel carrier **29** to prevent the torsion spring **124h** from sliding off the load wheel axle **30** or **31**. In the embodiment shown in FIGS. **18C** and **18D**, the torsion spring **124h** is positioned between the wheel carrier **29** and the load wheel **15** (or **14**) so the loop **126h** of the torsion spring **124h** cannot slide off of the load wheel axle **31** (or **30**).

(165) The torsion spring **124h** may include two idler arms **128h** that cross each other and are directly or indirectly connected to the separate idler wheels **116**, such as to axles (not shown) of the idler wheels **116**. Alternatively, the idler arms **128h** may have auxiliary projections **188** that insert into the axles of the idler wheels **116** or that function as the axles of the idler wheels **116**. In an alternative embodiment, the idler arms **128h** may slide through an axle slot (not shown, but such as shown in FIG. **12C**) in an idler axle (not shown, but such as shown in FIG. **12C**) of the idler wheel **116** of the torque-coupling assembly **78h** to provide force against the idler axle so that the force-applying coupler **122h** urges the idler wheel **116** against the forward load wheel **14** and the rear load wheel **15**. As noted previously, such a slot **138** may prevent rotation of the idler axle **121** in embodiments in which the idler wheel **116** constitutes a roller with bearings over the idler axle **121**.

(166) Typically, the torque-coupling assembly **78h** employs a torsion spring **124h** on each side of the wheel carrier **29**. The torsion springs **124h** can be employed around opposite sides of the same load wheel axle, or a first torsion spring **124h** can be employed around the rear axle **31** one side of the wheel carrier **29** while a second torsion spring **124h** can be employed around the forward axle **30** on the other side of the wheel carrier **29**. One will appreciate that torsion springs **124h** can be deployed around both the forward and rear axles **30** and **31** on both sides of the wheel carrier **29**.

(167) The torsion springs **124h** urge the top and bottom idler wheels **116** toward each other so that they contact the respective the upper surfaces **174** and lower surfaces **176** of the load wheels **14** and **15** and couple the torque of the forward load wheel **14** and the rear load wheel **15**. The strength of the torsion spring **124h** and/or the diameter **114** of the idler wheel **116** can be adjusted to determine the relative elevation of the idler wheel surfaces with respect to the surfaces of the forward load wheel **14** and the rear load wheel **15**. One will appreciate that the torque-coupling assembly **78h** can be adapted so that the bottom of the bottom idler wheel **116** is at or above the bottom of the forward load wheel **14** and the rear load wheel **15**.

(168) FIG. **19A** is a top plan view of a load wheel assembly **78i** employing another alternative embodiment of a torque-coupling assembly **78i**, and FIG. **19B** is a side elevation view of the load wheel assembly **78i** of FIG. **19A** with a wheel carrier strut **22** and a portion of a wheel carrier bracket **86** removed for clarity. The alternative torque-coupling assembly **78i** employs an idler gear wheel **116i** connected through bearings **191** to the wheel carrier **29**. Alternatively, the torque-coupling assembly **78i** employs idler gear wheels **116i** connected through bearings **191** on opposite sides of the wheel carrier **29** and that may be identical or different. Differences might include the idler wheel gear diameter. Each idler gear wheel **116i** interacts with one or more gears **192** that are fixed to the forward load wheel **14** and the rear load wheel **15** to transfer torque between them. Load wheel motion arrows **194** show the direction of rotation of the load wheels (and idler wheel motion arrow **196** shows the direction of motion of the idler gear wheel **116i**) when the wheel carrier **29** moves backward. These arrows would be reversed when the wheel carrier **29** moves in the forward direction.

(169) FIG. **20A** is a top plan view of a load wheel assembly **78j** employing another alternative embodiment of a torque-coupling assembly **78j**, and FIG. **20B** is a side elevation view of the load wheel assembly **78j** of FIG. **20A** with a wheel carrier strut **22** and a portion of a wheel carrier bracket **86** removed for clarity. The alternative torque-coupling assembly **78j** employs a “raceway” **198** connected to the wheel carrier **29**. Alternatively, the torque-coupling assembly **78j** employs raceways **198**, which may be identical or different, connected on opposite sides of the wheel carrier **29**. Each raceway **198** interacts with one or more gears **192j** that may be fixed to the forward load wheel **14** and the rear load wheel **15** to transfer torque between them.

(170) This embodiment is based on the recirculating ball concept (also known as worm and sector or recirculating ball and nut), such as commonly used in steering systems and ball screws. However, in this embodiment, no screw is involved. The ball bearings are simply used to form the torque transfer device using a formed gear to push the balls along a raceway to the driven gear. The pushing force provides the torque transfer.

(171) FIG. 21A is a top plan view of a load wheel assembly **78k** employing another alternative embodiment of a torque-coupling assembly **78k**, and FIG. 21B is a side elevation view of the load wheel assembly **78k** of FIG. 21A with a wheel carrier strut **22** and a portion of a wheel carrier bracket **86** removed for clarity. The alternative torque-coupling assembly **78k** employs an idler wheel **116k** that has an idler wheel axle **121k** that is aligned with the carrier axle **27k**. Moreover, the carrier axle **27k** and the idler wheel axle **121k** may be a single component. One or both of the forward load wheel **14** and the rear load wheel **15** can be a smaller size than in the other embodiments, or the forward load wheel **14** and the rear load wheel **15** can be spaced further apart than in other embodiments to accommodate the diameter **114** of the idler wheel **116k**. The diameter **114** of the idler wheel **116k** and the material of the idler wheel **116k** can be adapted to couple torque between the forward load wheel **14** and the rear load wheel **15**. For example, the material of the idler wheel **116k** may comprise an elastic material such as neoprene, polyurethane or Santoprene™ and may include longitudinal apertures **117k** that facilitate the idler wheel **116k** compressing when contacted by the surfaces of the load wheels **17**. The stiffness of the material for the idler wheel **116k** and the size and shape of the apertures **117k** may be designed to increase or decrease the amount of pressure exerted by the idler wheel **116k** on the forward load wheel **14** and the rear load wheel **15**.

(172) FIG. 22A is a top plan view of a load wheel assembly **78m** employing another alternative embodiment of a torque-coupling assembly **78m**, and FIG. 22B is a side elevation view of the load wheel assembly **78m** of FIG. 22A with a wheel carrier strut **22** and a portion of a wheel carrier bracket **86** removed for clarity. The alternative torque-coupling assembly **78m** employs one or more toothed belts **204** that may be on only one side of the wheel carrier **29** or on opposite sides of the wheel carrier **29**. Each toothed belt **204** interacts with one or more gears or pulleys **192m** that may be fixed to the forward load wheel **14** and the rear load wheel **15** to transfer torque between them.

(173) FIG. 22C is a top plan view of a load wheel assembly **78n** that is similar to that shown in FIG. 22A but additionally comprising a lead-in roller **212** in the form of a paddle wheel having multiple paddles **214**. FIG. 22D is a first side elevation view of the load wheel assembly **78** with a wheel carrier strut **22** and a portion of a wheel carrier bracket **86** removed for clarity, and FIG. 22E is a second side elevation view of the load wheel assembly **78n** with a wheel carrier strut **22** and a portion of a wheel carrier bracket **86** removed for clarity. FIGS. 22D and 22E are views from opposite sides of the load wheel assembly **78n**.

(174) The lead-in roller **212** is shown in the form of a paddle wheel having paddles **214**; however, any type of wheel can be employed. By way of illustration and not limitation, the lead-in roller **212** may have, instead of paddles **214**, other features on its surface, such as ridges, bumps, or other surface texture to enable it to grip and climb over an obstacle, such as a base board **2**, on the floor, rather than push it horizontally. As another example, the lead-in roller **212** may have a tacky outer surface, such as a rubber tire; alternatively, the entire lead-in roller may be made from a solid tacky material, such as rubber, a rubber-based compound, or a rubber-like material. The effective diameter of the lead-in roller **212** may be different from that of the load wheels **17**. For example, the effective diameter of the lead-in roller **212** may be less than or equal to that of the load wheels **17**, as the lead-in roller is preferably not load bearing like the load wheels **17**; instead, the lead-in roller **212** is preferably meant to climb over an obstacle, such as a base board **2**, on the floor but otherwise not contact a flat floor. To that end, the lead-in roller may have the same or different (larger or smaller) diameter as the load wheels **17** but positioned upward in a horizontally biased position relative to the load wheels **17**. That may be accomplished, for example, by having bent wheel carriers **29** (e.g., right end up in FIG. 22D, left end up in FIG. 22E), instead of the straight ones illustrated in FIGS. 22D and 22E.

(175) The lead-in roller **212** may be coupled to the torque of one or more of the load wheels **17** through a torque-coupling assembly **78n**, which may employ, for example, a toothed belt **216** that interacts with a gear **192n** connected to an axle **218** of the lead-in roller **212**, as shown in FIGS.

22C-22E. One will appreciate that the torque-coupling assembly **78** may be employed to couple the lead-in roller **212** with only the rear load wheel **15**, with only the forward load wheel **14**, or with both of the rear load wheel **15** and the forward load wheel **14**. One will further appreciate that the torque-coupling assembly **78n** may be employed on only one side of the load wheel unit, as shown, or with both sides of the load wheel unit **78**.

(176) One will also appreciate that torque coupling between the lead-in roller **212** and one or both of the load wheels **17** can be implemented in any manner, such as by any of the force applying couplers **122a-122h**. Other examples of mechanism to couple torque to the lead-in roller **212** include, for example, chain(s) and gears or untoothed belt(s). Similarly, the lead-in roller **212** may be utilized in conjunction with any other type of load wheel assembly, such as any of the load wheel assemblies **78-78k**, or with load wheel assemblies that do not couple torque between their load wheels. Alternatively, the lead-in roller **212** may not be coupled to either of the load wheels **17**.

(177) Load wheels with torque-coupling assemblies may be used in conventional, non-modular, load wheel arrangements or other suitable environments as well.

(178) While some of the examples have been illustrated or described with respect to providing functionality for a “walkie” or “rider” style pallet truck, some or all of the features may also be enabled for operation with other types of industrial vehicles including, but not limited to, reach trucks, three-wheel stand trucks, warehouse trucks, and counterbalanced trucks.

CONCLUSION

(179) The terms and descriptions used above are set forth by way of illustration and example only and are not meant as limitations. Those skilled in the art will recognize that many variations, enhancements and modifications of the concepts described herein are possible without departing from the underlying principles of the invention. For example, skilled persons will appreciate that the subject matter of any sentence or paragraph can be combined with subject matter of some or all of the other sentences or paragraphs, except where such combinations are mutually exclusive. The scope of the invention should therefore be determined only by the following claims, claims presented in a continuation patent application, and equivalents to the foregoing claims.

Claims

1. A fork assembly for a forked material-handling vehicle configured for lifting a pallet having one or more upper planks, the fork assembly comprising: a discrete elongate body having an elongate body upper surface; a discrete load wheel module having a load wheel frame and a load wheel module upper surface and one or more load wheels, wherein the load wheel module is configured for moving the one or more load wheels between an undeployed position and a deployed position, wherein the one or more load wheels in the undeployed position are at least partly surrounded by the load wheel frame, wherein the load wheel module is configured for fitting beneath the one or more upper planks of a pallet when the one or more load wheels are in the undeployed position, and wherein a majority of a wheel height of the one or more load wheels extends beyond the load wheel frame in the deployed position; a proximal interlocking mechanism detachably connecting the elongate body to the load wheel module; a discrete fork tip configured for fitting beneath the one or more upper planks of a pallet; and a distal interlocking mechanism detachably connecting the load wheel module to the fork tip, wherein the load wheel module is positioned between the elongate body and the fork tip, and wherein the elongate body upper surface and the load wheel module upper surface are substantially coplanar.

2. The fork assembly of claim 1, wherein the load wheel module is configured to house the majority of the wheel height of one or more load wheels within the load wheel frame when the one or more load wheels are in the undeployed position.

3. The fork assembly of claim 1, wherein the load wheel module is configured to extend the wheel

height of the one or more load wheels beyond the load wheel frame when the one or more load wheels are in the deployed position.

4. The fork assembly of claim 1, wherein the pallet comprises a base board spaced apart from the one or more upper planks, and wherein the fork tip is configured to fit between the base board and the one or more upper planks of the pallet.
5. The fork assembly of claim 1, wherein the pallet comprises a base board spaced apart from the one or more upper planks, and wherein the load wheel module is configured to fit between the base board and the one or more upper planks of the pallet when the one or more load wheels in the undeployed position.
6. The fork assembly of claim 1, wherein at least one of the proximal interlocking mechanism and the distal interlocking mechanism employs mated shear-resistant features, including first and second shear-resistant features.
7. The fork assembly of claim 6, wherein the shear-resistant features are configured to receive a fastener.
8. The fork assembly of claim 1, wherein at least one of the proximal interlocking mechanism and the distal interlocking mechanism employs a mated internally threaded receptacle and an externally threaded fastener.
9. The fork assembly of claim 1, wherein the proximal interlocking mechanism employs a proximal internally threaded receptacle that is mated to a proximal externally threaded fastener, wherein one of the proximal internally threaded receptacle and the proximal externally threaded fastener is configured to connect to the elongate body, and wherein a different one of the proximal internally threaded receptacle and the proximal externally threaded fastener is configured to connect to the load wheel module.
10. The fork assembly of claim 1, wherein the proximal interlocking mechanism includes a proximal protruding feature that is mated to a proximal receiving feature, wherein one of the proximal protruding feature and the proximal receiving feature is connected to the elongate body, wherein a different one of the proximal protruding feature and the proximal receiving feature is connected to the load wheel module.
11. The fork assembly of claim 10, wherein both of the proximal protruding feature and the proximal receiving feature are configured to receive a proximal externally threaded fastener.
12. The fork assembly of claim 1, wherein the proximal interlocking mechanism includes a proximal endcap that is attached to the elongate body and includes a proximal shear-resistant body feature that is mated to a proximal shear-resistant module feature of the load wheel module.
13. The fork assembly of claim 12, wherein the proximal endcap is welded to the elongate body.
14. The fork assembly of claim 12, wherein the distal interlocking mechanism includes a distal endcap that is attached to the fork tip and includes a distal shear-resistant attachment feature that is mated to a distal shear-resistant module feature of the load wheel module, wherein the proximal and distal interlocking mechanisms are operatively identical.
15. The fork assembly of claim 12, wherein the distal interlocking mechanism includes a distal endcap that is attached to the fork tip and includes a distal shear-resistant attachment feature that is mated to a distal shear-resistant module feature of the load wheel module.
16. The fork assembly of claim 15, wherein the distal shear-resistant attachment feature and the distal shear-resistant module feature are adapted to receive a fastener.
17. The fork assembly of claim 1, wherein the elongate body has a first characterizing color, wherein the load wheel module has a second characterizing color, wherein the fork tip has a third characterizing color, and wherein the first, second, and third characterizing colors are different.
18. The fork assembly of claim 1, wherein the one or more load wheels comprise a forward load wheel and a non-motorized rear load wheel contained by a load wheel assembly, and wherein a non-motorized torque-coupling assembly is connected between the forward load wheel and the rear load wheel.

19. The fork assembly of claim 1, wherein the fork tip includes a proximal connection end for attachment closest to the load wheel module, wherein the proximal connection end has proximal end dimensions, wherein the fork tip includes a distal end having distal end dimensions, and wherein at least one of the tip distal end dimensions is smaller than a respective one of the proximal end dimensions.
20. The fork assembly of claim 1, wherein the load wheel module further comprises a hydraulic actuator contained within the load wheel frame and operatively connected to deploy the one or more load wheels hydraulically.
21. The fork assembly of claim 1, wherein the forked material-handling vehicle comprises a pallet truck.
22. The fork assembly of claim 1, wherein the load wheel module employs a pivotal connection between the load wheel frame and the one or more load wheels.
23. A pallet truck including a fork assembly configured for lifting a pallet having one or more upper planks, the pallet truck comprising: a steer wheel; a chassis operatively connected to the steer wheel; and two substantially parallel first and second fork assemblies operatively connected to and extending from the chassis and configured to hold a load for conveyance by the pallet truck as the pallet truck moves; wherein the first fork assembly comprises: a first discrete elongate body having a first elongate body upper surface; a first discrete load wheel module having a first load wheel frame and a first load wheel module upper surface and one or more first load wheels, wherein the first load wheel module is configured for moving the one or more first load wheels between an first undeployed position and a first deployed position, wherein the one or more first load wheels in the first undeployed position are at least partly surrounded by the first load wheel frame, wherein the first load wheel module is configured for fitting beneath the one or more upper planks of a pallet when the one or more first load wheels are in the first undeployed position, and wherein a first majority of a first wheel height of the one or more first load wheels extends beyond the first load wheel frame in the first deployed position; a first proximal interlocking mechanism detachably connecting the first elongate body to the first load wheel module; a discrete first fork tip configured for fitting beneath the one or more upper planks of a pallet; and a first distal interlocking mechanism detachably connecting the first load wheel module to the first fork tip, wherein the first load wheel module is positioned between the first elongate body and the first fork tip, and wherein the first elongate body upper surface and the first load wheel module upper surface are substantially coplanar; and wherein the second fork assembly comprises: a second discrete elongate body having an second elongate body upper surface; a second discrete load wheel module having a second load wheel frame and a second load wheel module upper surface and one or more second load wheels, wherein the second load wheel module is configured for moving the one or more second load wheels between an second undeployed position and a second deployed position, wherein the one or more second load wheels in the second undeployed position are at least partly surrounded by the second load wheel frame, wherein the second load wheel module is configured for fitting beneath the one or more upper planks of a pallet when the one or more second load wheels are in the second undeployed position, and wherein a second majority of a second wheel height of the one or more second load wheels extends beyond the second load wheel frame in the second deployed position; a second proximal interlocking mechanism detachably connecting the second elongate body to the second load wheel module; a discrete second fork tip configured for fitting beneath the one or more upper planks of a pallet; and a second distal interlocking mechanism detachably connecting the second load wheel module to the second fork tip, wherein the second load wheel module is positioned between the second elongate body and the second fork tip, and wherein the second elongate body upper surface and the second load wheel module upper surface are substantially coplanar.
24. The pallet truck of claim 23, wherein the first load wheel module is configured to house the first majority of the first wheel height of one or more first load wheels within the first load wheel

frame when the one or more first load wheels are in the first undeployed position.

25. The pallet truck of claim 23, wherein the first load wheel module is configured to extend the first wheel height of the one or more first load wheels beyond the first load wheel frame when the one or more first load wheels are in the first deployed position.

26. The pallet truck of claim 23, wherein the pallet comprises a base board spaced apart from the one or more upper planks, and wherein the first fork tip is configured to fit between the base board and the one or more upper planks of the pallet.

27. The pallet truck of claim 23, wherein the pallet comprises a base board spaced apart from the one or more upper planks, and wherein the first load wheel module is configured to fit between the base board and the one or more upper planks of the pallet when the one or more first load wheels in the first undeployed position.

28. The pallet truck of claim 23, wherein the first and second load wheel modules are interchangeable, wherein the first and second fork tips are interchangeable, wherein the first and second proximal interlocking mechanisms are interchangeable, and wherein the first and second distal interlocking mechanisms are interchangeable.

29. The pallet truck of claim 23, wherein the first and second fork assemblies are spaced-apart and have unobstructed distinct distal ends.

30. The pallet truck of claim 23, further comprising: a hydraulic power source; and a first hydraulic line positioned through the first elongate body, wherein the first hydraulic line transmits hydraulic fluid from the hydraulic power source to a first hydraulic actuator positioned completely within the first load wheel module.

31. The pallet truck of claim 23, wherein the first proximal interlocking mechanism includes a first endcap that is attached to the first elongate body and includes a first sheer-resistant body feature that is mated to a first sheer-resistant module feature of the first load wheel module.

32. The pallet truck of claim 23, wherein the first proximal interlocking mechanism includes a first protruding feature that is mated to a first receiving feature, wherein one of the first protruding feature and the first receiving feature is permanently connected to the first elongate body, wherein a different one of the first protruding feature and the first receiving feature is permanently connected to the first load wheel module.

33. The pallet truck of claim 23, wherein the first and second elongate bodies have a first characterizing color, wherein the first and second load wheel modules have a second characterizing color, wherein the first and second fork tips have a third characterizing color, and wherein the first, second, and third characterizing colors are different.

34. An inventory of parts for a forked material-handling vehicle configured for lifting a pallet having one or more upper planks, the inventory comprising: multiple interchangeable elongate bodies having respective elongate body upper surfaces, wherein the multiple interchangeable elongate bodies include a first elongate body having a first elongate body upper surface and a second elongate body having a second elongate body upper surface; multiple interchangeable load wheel modules having respective load wheel module upper surfaces, wherein the multiple interchangeable load wheel modules include a first load wheel module and a second load wheel module, wherein the first load wheel module has a first load wheel frame and a first load wheel module upper surface and one or more first load wheels, wherein the first load wheel module is configured for moving the one or more first load wheels between an first undeployed position and a first deployed position, wherein the one or more first load wheels in the first undeployed position are at least partly surrounded by the first load wheel frame, wherein the first load wheel module is configured for fitting beneath the one or more upper planks of a pallet when the one or more first load wheels are in the first undeployed position, wherein a first majority of a first wheel height of the one or more first load wheels extends beyond the first load wheel frame in the first deployed position, wherein the second load wheel module has a second load wheel frame and a second load wheel module upper surface and one or more second load wheels, wherein the second load wheel

module is configured for moving the one or more second load wheels between an second undeployed position and a second deployed position, wherein the one or more second load wheels in the second undeployed position are at least partly surrounded by the second load wheel frame, wherein the second load wheel module is configured for fitting beneath the one or more upper planks of a pallet when the one or more second load wheels are in the second undeployed position, and wherein a second majority of a second wheel height of the one or more second load wheels extends beyond the second load wheel frame in the second deployed position; multiple interchangeable proximal interlocking mechanism components configured for detachable connection of any one of the multiple interchangeable elongate bodies to any one of the multiple interchangeable load wheel modules such that the respective elongate body upper surface and the respective load wheel module upper surface are aligned to be substantially coplanar; multiple interchangeable fork tips, including a first fork tip and a second fork tip, wherein the first fork tip and second for tip are configured for fitting beneath the one or more upper planks of a pallet; and multiple distal interlocking proximal mechanism components configured for detachable connection of any one of the multiple interchangeable fork tips to any one of the multiple interchangeable load wheel modules.

35. The inventory of parts of claim 34, wherein the multiple proximal or distal interchangeable interlocking mechanism components include mated shear-resistant features, including first sheer-resistant features and second sheer-resistant features.

36. The inventory of parts of claim 35, wherein the mated sheer-resistant features are configured to receive a fastener.

37. The inventory of parts of claim 34, wherein the multiple proximal or distal interchangeable interlocking mechanism components employ a mated internally threaded receptacle and an externally threaded fastener.

38. The inventory of parts of claim 34, wherein the multiple proximal or distal interchangeable interlocking mechanism components include a protruding feature that is mated to a receiving feature, wherein one of the protruding feature and the receiving feature is permanently connected to a respective elongate body, wherein a different one of the protruding feature and the receiving feature is permanently connected to the respective load wheel module.

39. The inventory of parts of claim 34, wherein the first and second elongate bodies have a first characterizing color, wherein the first and second load wheel modules have a second characterizing color, wherein the first and second fork tips have a third characterizing color, and wherein the first, second, and third characterizing colors are different.

40. The inventory of parts of claim 34, wherein the first and second elongate bodies have equal lengths, and wherein the multiple interchangeable elongate bodies further comprise multiple third elongate bodies having a different length than that of the first and second elongate bodies.

41. The inventory of parts of claim 34, wherein the first and second load wheel modules are of the same style, wherein the inventory of parts further comprises multiple third load wheel modules having a different style than that of the first and second load wheel modules.

42. The inventory of parts of claim 34, wherein the load wheel module further comprises a hydraulic actuator contained within the load wheel frame and operatively connected to deploy the one or more load wheels hydraulically.
